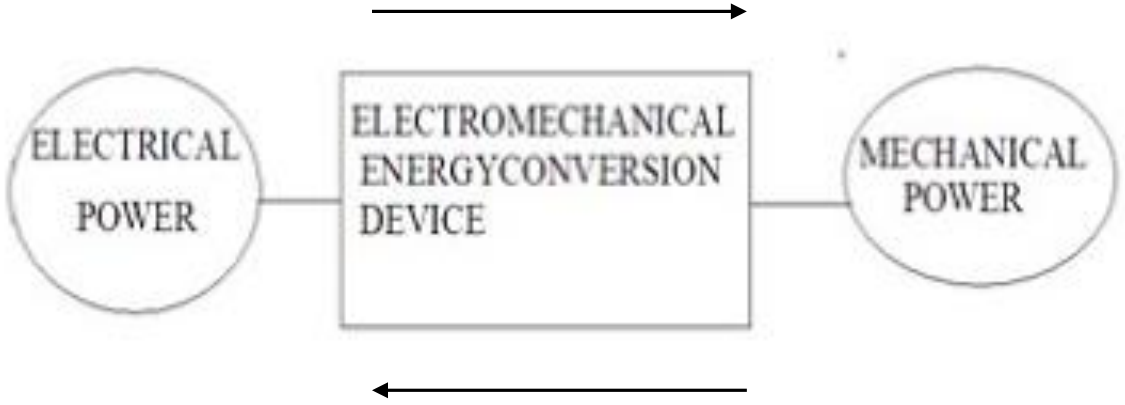
UNIT-5 Electrical Machines

Introduction

Electric Machine: An electrical machine converts electrical energy into mechanical energy & vice versa.

A transformer is also an Electrical machine with an exception that it converts voltage & current levels.

The process of translation is known as Electromechanical energy conversion.



Electric generator: An **electric generator** is an electrical machine which converts mechanical energy into electrical energy.

- A generator works on the principle of electromagnetic induction.It states that whenever a conductor moves in a magnetic field, an emf gets induced within the conductor.

These are two types:
1.AC Generator
2.DC Generator

Electric motor: An electric motor is an electrical machine which converts electrical energy into mechanical energy.

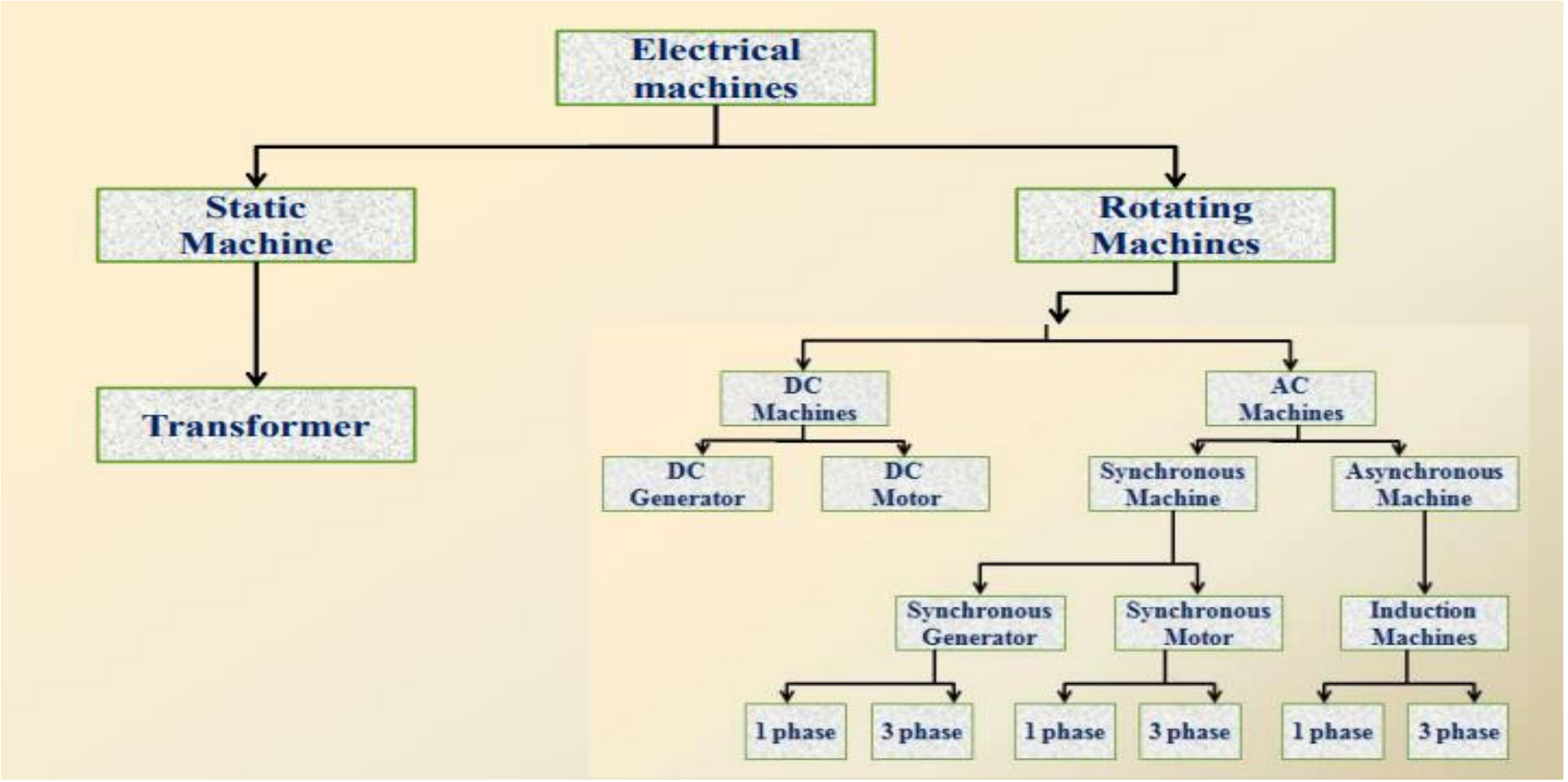
- When a current carrying conductor is placed in a magnetic field, the conductor experiences a mechanical force and this is the principle behind motoring action.

These are two types:
1.AC motors: (i)Induction motors and (ii) Synchronous motor
2.DC motors: (i)Brushed DC motor and (ii) Brushless DC motor

Transformer: Transformers do not actually make conversion between mechanical and electrical energy, but they transfer electric power from one circuit to another circuit. They can increase or decrease (step-up or step-down) the voltage while transferring the power without changing the frequency, but with the corresponding decrease or increase in the current.

- Transformer works on the principle mutual induction.

Classification of Electrical machine



DC machines

The DC machine can be classified into two types namely DC motors **as well as DC** generators.

The output of the DC machine is DC output because they convert AC voltage to DC voltage.

The conversion of this mechanism is known as the commutator, thus these machines are also named as commutating machines.

The **applications of the DC machine** is limited to trains, mills, and mines. For example, underground subway cars, as well as trolleys, may utilize DC motors.

1.DC Generator: If the Direct current is used to convert mechanical energy in to electrical energy the machine is called D.C.Generator.

2.DC Motor: If the Direct current is used to convert electrical energy in to mechanical energy the machine is called D.C.Motor.

DC Motor Working principle

If the Direct current is used to convert electrical energy in to mechanical energy the machine is called D.C. Motor.

Working principle : DC motors operate on Faraday's principle of electromagnetism which states that a current-carrying conductor experiences a force when placed in a magnetic field. The experienced force is called the Lorentz force. The Fleming left-hand rule gives the direction of the force.

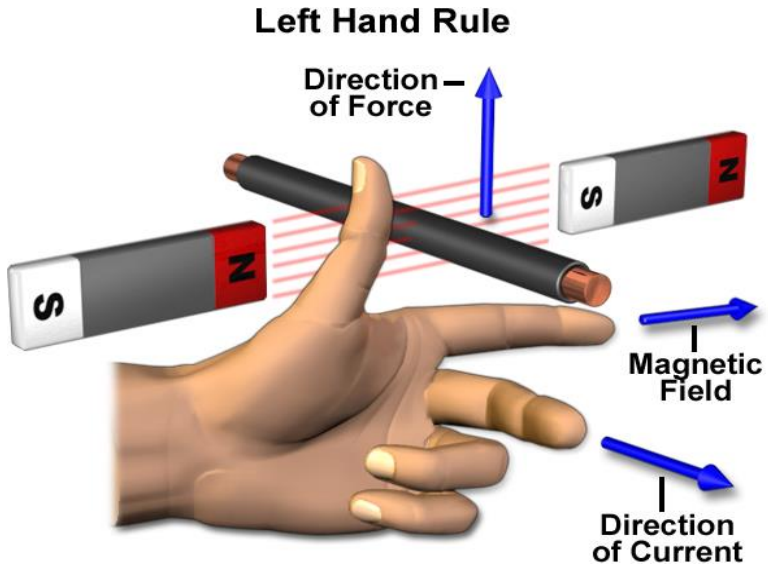
F = BIL

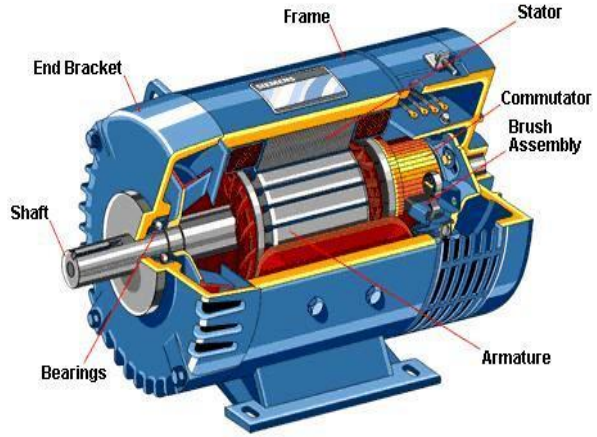
where F is force,

B is the magnetic field,

I stand for current,

L is the length of the conductor).





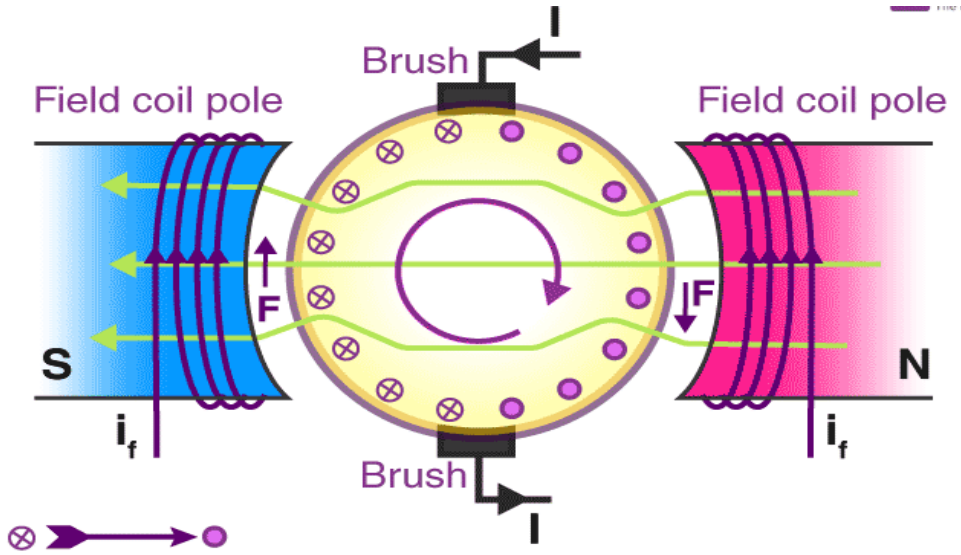
Working of DC motor

The magnetic field is setup by the field winding.

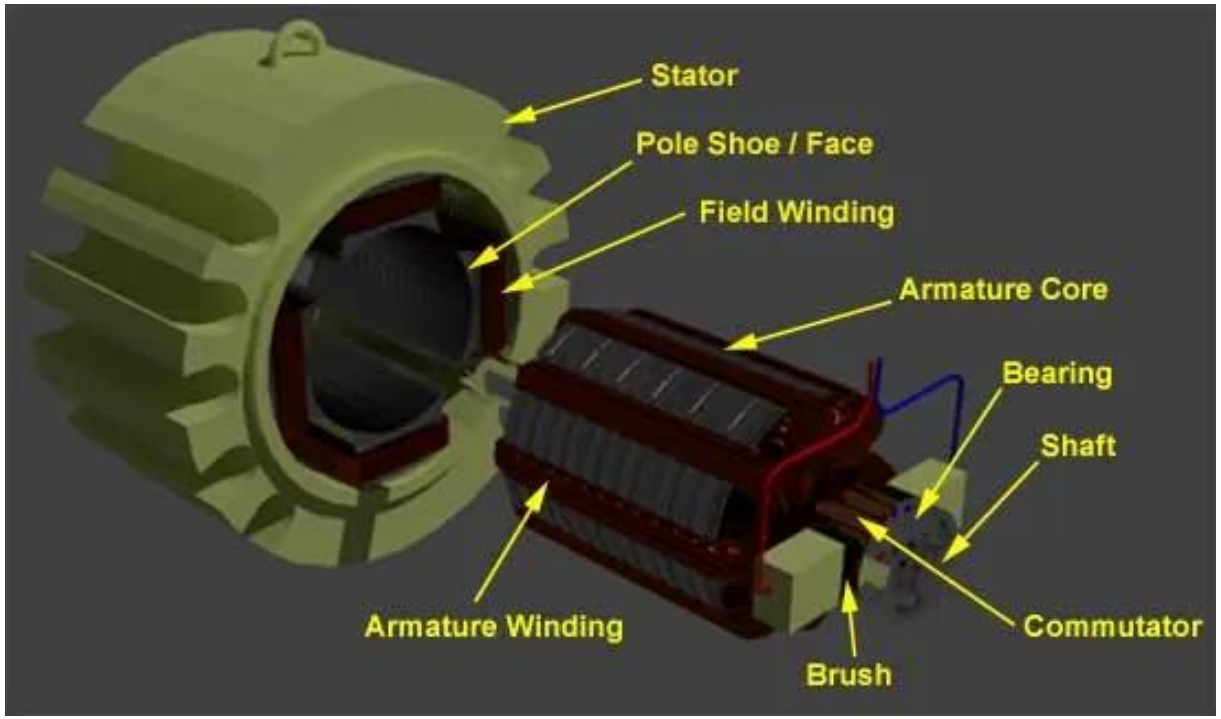
When the supply is given to armature conductors, the current will flow through the conductors. It experience mechanical force and starts to rotate. Current direction of conductors under N-pole downwards and under S-pole upwards.

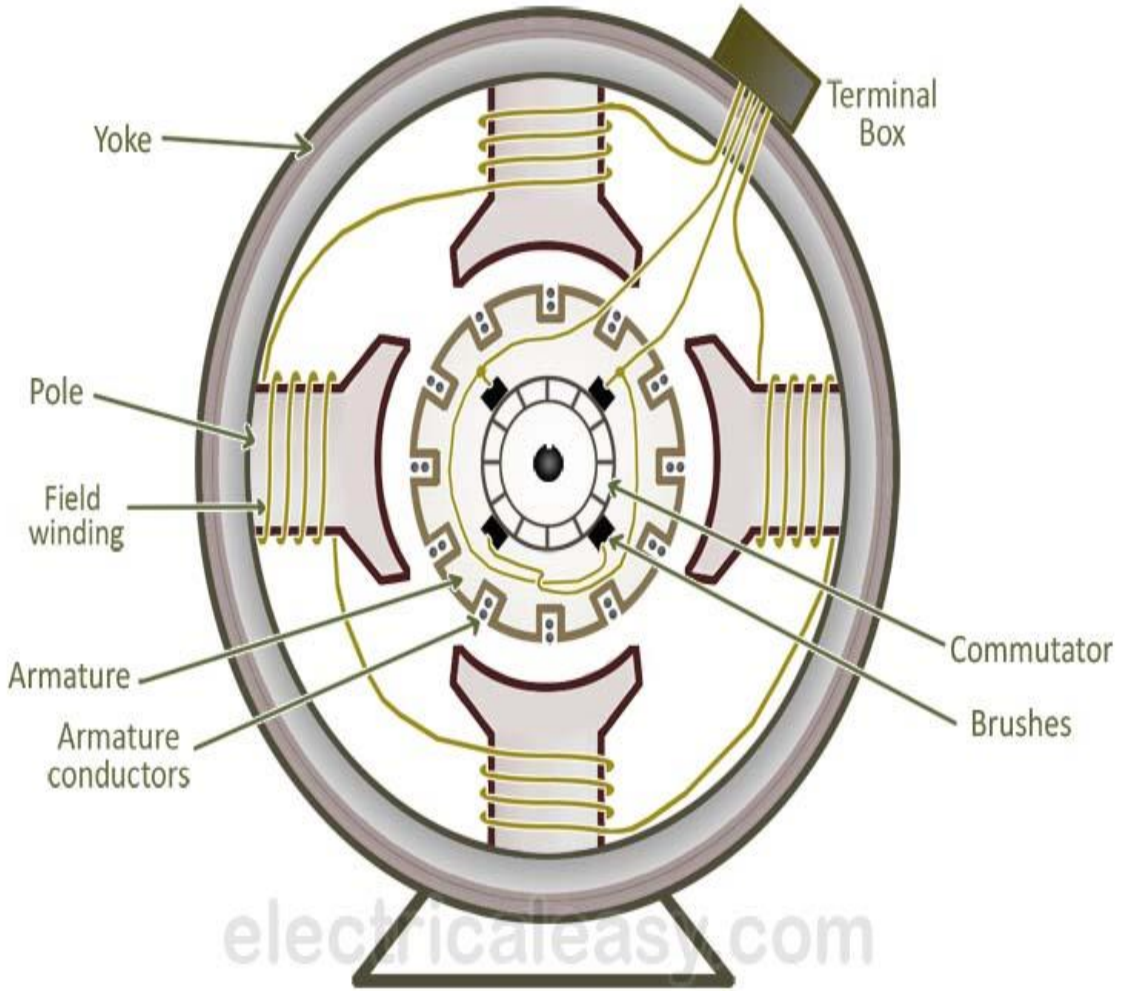
This direction obtained by Fleming's left hand rule. each conductor experiences a force and it tends to rotate the armature in anticlockwise direction. It produce driving torque which makes the armature rotating.

Note : The construction of DC generator is same as that of the DC motor. It means, a DC machine can be used both as a generator and as a motor.



Construction of DC Motor





Construction of DC Generator

The major components of a DC machine are:

1.Magnetic frame or yoke

2.Pole core and pole shoe

3.Field coil or winding

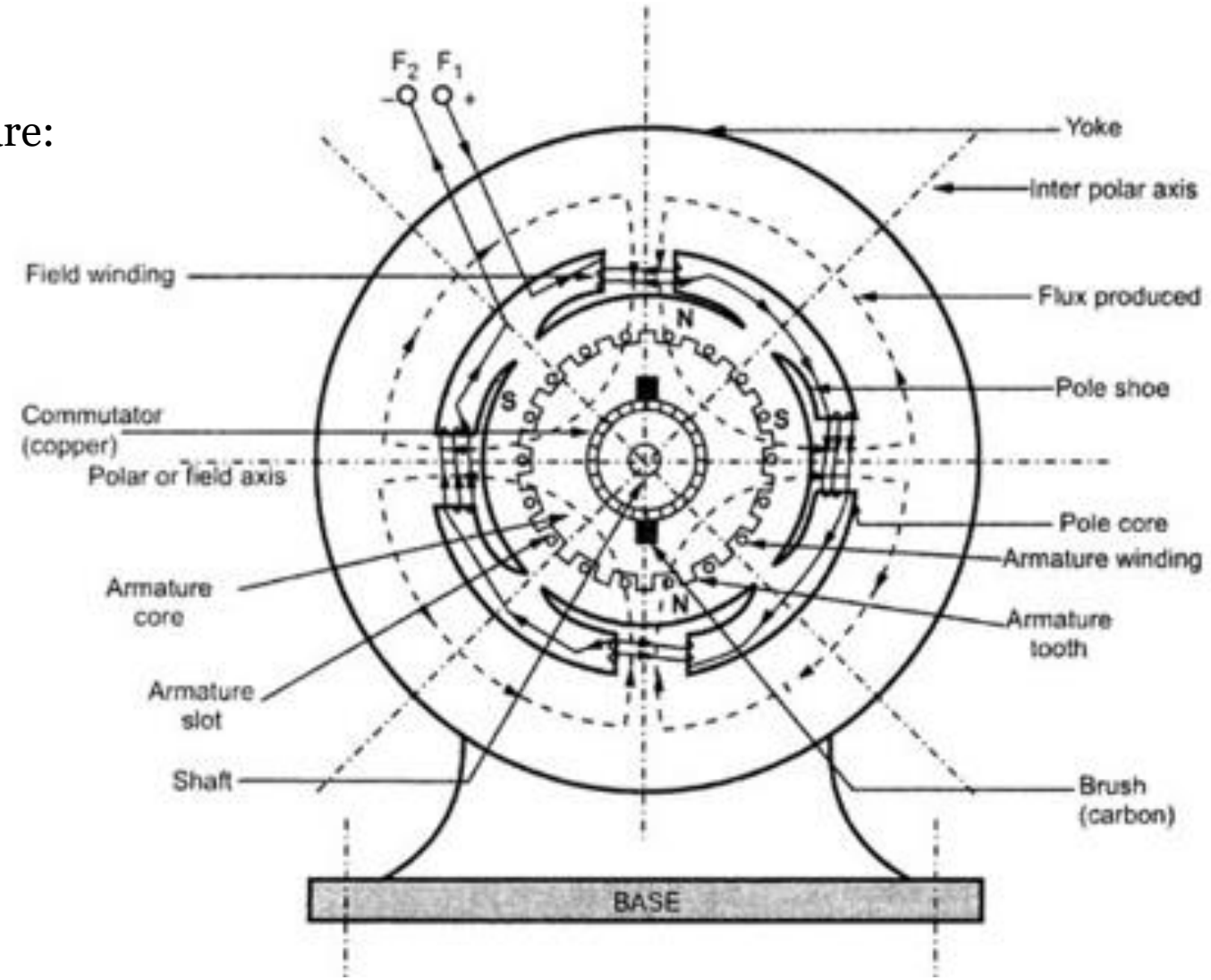
4.Armature core and winding

5.Commutator

6.Brushes

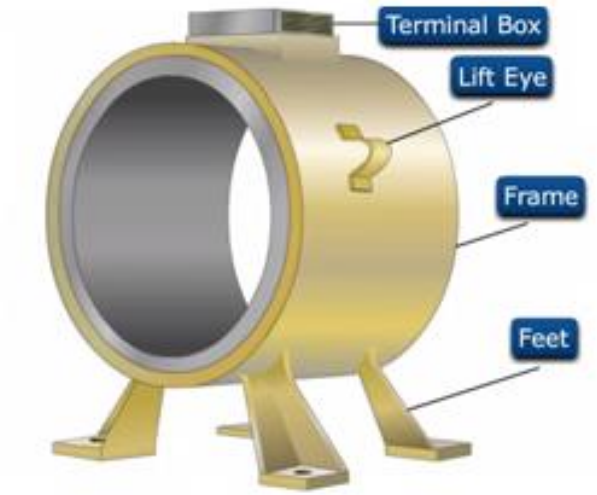
7.Bearings and shaft

8.Terminal box



1. Magnetic frame or yoke

- The outermost cylindrical frame is called yoke. It provides protection to the rotating and other part of the machine from moisture, dust etc
- Yoke is usually made of cast iron for small machine, because of its cheapness.
- But for large machines, it is made of cast steel or rolled steel, due to its high permeability.
- It is used to carry the magnetic flux produced by the poles. It provides low reluctance path for the magnetic flux.
- It provides mechanical support for the poles.
- The lifting eye, feet and the terminal box are welded to the frame afterwards.



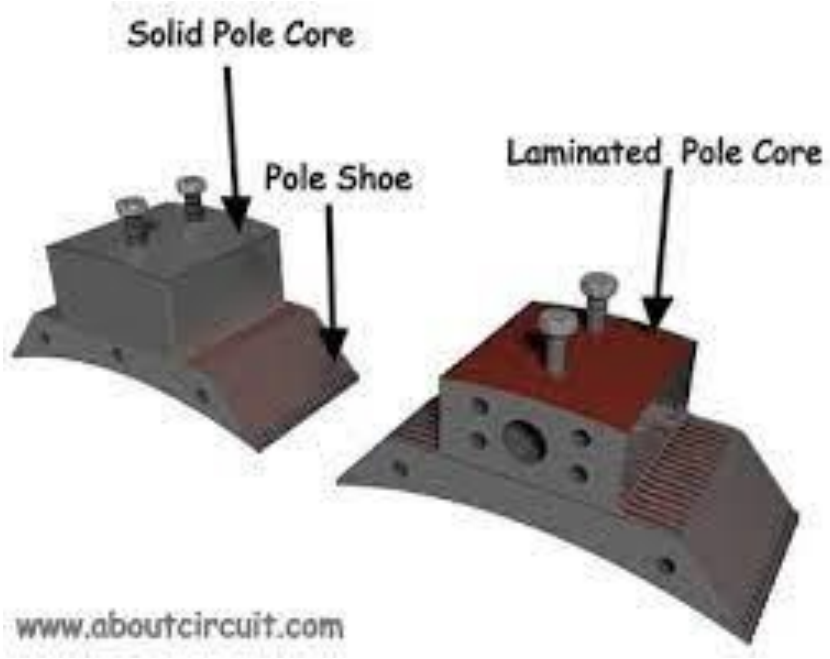
2.Pole core and pole shoe

- The field pole consists of pole cores, pole shoes and field winding.

- The poles are made of thin laminated sheets, to avoid heating and eddy current loss.

- **Pole cores:** These are the projecting rectangular parts, which produce magnetic flux needed for the generator, when it is excited by the field winding. It is fitted to the yoke or frame by means of bolts and nuts or rivets.

Pole shoes : These are located at the end of pole core. It makes magnetic field uniform on the surface of the armature. Due to its typical shape, it enlarges the area of the pole, so that more flux can pass through the air gap to armature. Cast iron or Cast steel is used.



Following are the main function of the poles

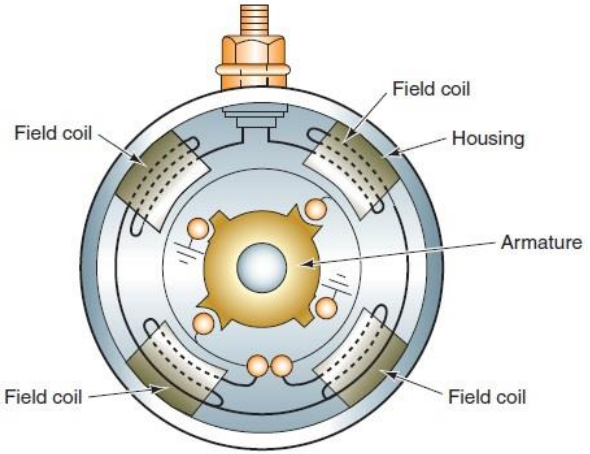
- it acts as a mechanical support to the field coil.
- they reduce the reluctance of the magnetic path.
- they guide and spread out the flux in the air gap.

3.Field coil or winding

- The coil wound on the pole core are called field coils. Field coils are connected in series to form field winding. Field winding is also called as Exciting winding.

- Current is passed through the field winding in a specific direction, to magnetize the poles and pole shoes. Thus magnetic flux is produce in the air gap between the pole shoe and armature.

- Field coil is made up of copper.
- Field winding is divided into various coils called field coils.
- These are connected in series with each other and in such a direction around pole cores, such that alternate 'N' and 'S' poles are formed.



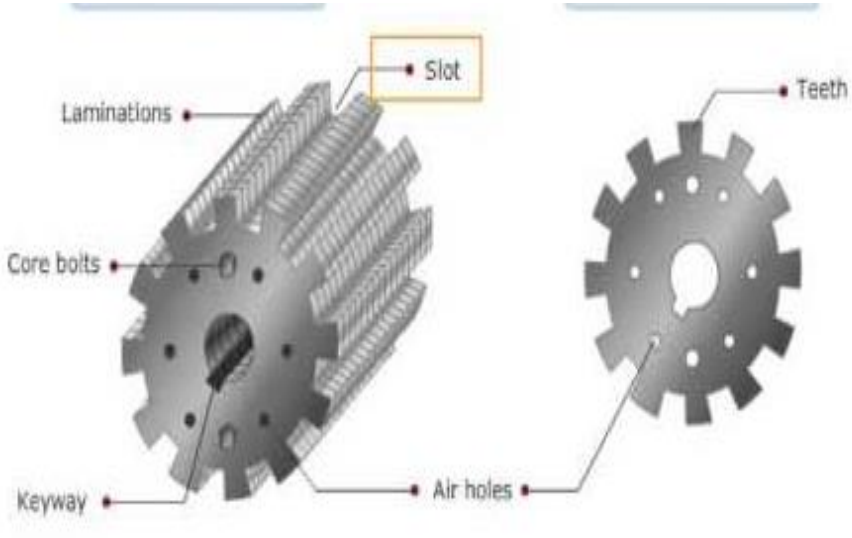
4.Armature core and winding

(i) Armature core: It is cylindrical in shape and made up of high Permeability and low reluctance cast steel or cast iron is used..The laminations are insulated by thin paper or thin coating of varnish, thickness is about 0.3 mm to 0.5mm.

- These lamination will reduce the eddy current loss..

Purpose:

- Armature core provides house for armature winding i.e. armature conductors
- To provide a path of low reluctance to the magnetic flux produced by the field winding.
- Due to losses, heat will be developed in the armature. To dissipate this heat, a fan is provided at one end of armature. Ventilating ducts (air holes) are also provided in the armature for the purpose of cooling. The width of the ventilating ducts varies from 5 to 10 mm.



(ii)Armature winding

- Armature conductor is placed in a armature slots present on the periphery of armature core.
- Armature conductor are interconnected to form the armature winding.
- When the armature winding is rotated using a prime mover, it cuts the magnetic flux lines and voltage gets induced in it.
- Armature winding is connected to the external circuit (load) through the commutator and brushes. It made up of copper.



Purpose:

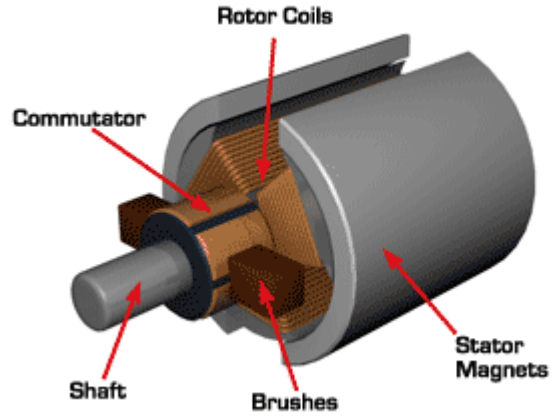
- Generation of e.m.f takes place in the armature winding in case of generators.
- To carry the current supplied in case of d.c. motors.
- To do the useful work in the external circuit.

Armature winding can be wound by one of the two methods; lap winding or wave winding.

Double layer lap or wave windings are generally used. A double layer winding means that each armature slot will carry two different coils.

5.Commutator

- The basic nature of e.m.f. induced in the armature conductors is alternating. This needs rectification in case of d.c. generator, which is possible by a device called commutator.
- It is cylindrical in shape and is made up of wedge shaped segments of the hard drawn, high conductivity copper.
- These segments are insulated from each other by thin layer of mica. Each commutator segment is connected to the armature
- conductor by means of copper lug or strip.



Purpose:

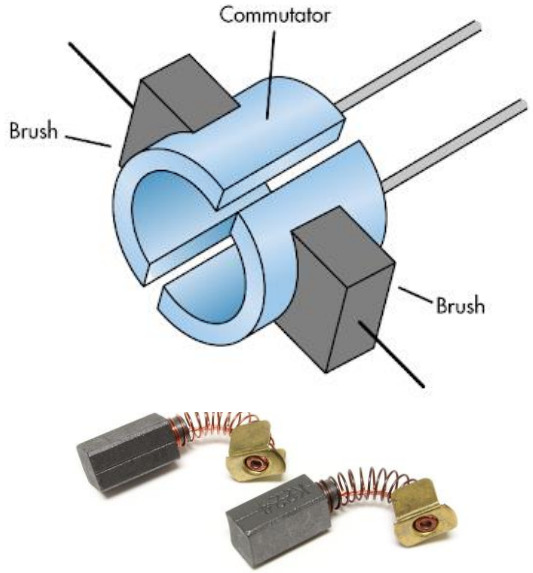
- To facilitate the collection of current from the armature conductors.
- To convert internally developed alternating e.m.f. to unidirectional (d.c.) e.m.f.
- To produce unidirectional torque in case of motors.

6.Brushes

- The brushes are made of high grade carbon because carbon is conducting material and at the same time in powdered form provides lubricating effect on the commutator surface.
- These are rectangular block shaped.

Purpose:

The main function of the brush is to collect current from commutator and supply it to the external load circuit.



Brushes

7.Bearings and shaft

(i)Bearings:

- For construction of smaller DC generator, ball bearings are used at both the ends of the shaft .
- Larger machines, roller bearings are used at the driving end and ball bearings are used at the non driving end of the machine.
- High carbon steel is used for the construction of bearings.

Purpose:

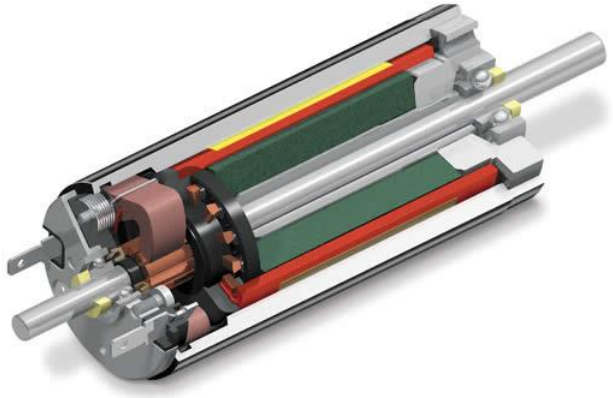
It reduce friction between the rotating and stationary parts of the machine.



Bearings

(ii)Shaft:

- The shaft is made up of mild steel having maximum breaking strength.
- It is used to transfer the mechanical power from or to the machine.
- All the rotating parts including the armature core, commutator, cooling parts and mounted and keyed to the shaft.





Shaft

8.Terminal box

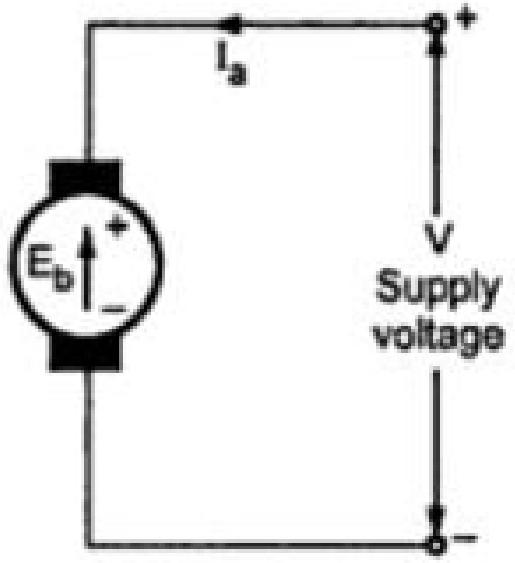
- This is an insulated box which carries the brass nuts and bolts to which wires from the brushes and field poles are brought out for connection with the external circuit.



Terminal box

E.m.f. Equation

- Let,
- ϕ = Flux per pole in Weber.
- Z = Total number of armature conductors.
- N = Armature rotation in revolution per minute (rpm).
- P = Number of poles.
- A = Number of parallel paths in armature.
- E = EMF induced in any parallel path or generated emf.



According to Faraday's law of Electromagnetic induction. Average emf generated per c

$$= \frac{d\phi}{dt} = \frac{flux\ cut}{time\ taken}\ volt$$

Flux cut by one conductor $= P\varphi$ wb ... .(1)

Time taken to complete one revolution is given as $t = \frac{60}{N}$ seconds ... .(2)

Therefore, the average induced e.m.f in one conductor will be: $e = \frac{P\varphi}{t}$ ... .(3)

Putting the value of (t) from Equation (2) in the equation (3) we will g $e = \frac{P\varphi}{60/N} = \frac{P\varphi N}{60}$ volts ... .(4)

The number of conductors connected in series in each parallel path = Z/A.
Therefore, the average induced e.m.f across each parallel path or the armature terminals is given by the equation shown below:

$$E = \frac{P\varphi N}{60} \times \frac{Z}{A} = \frac{PZ\varphi N}{60\,A} \text{ volts or}$$

$$E = \frac{PZ\varphi n}{A} \quad \ldots\ldots(5)$$

Where n is the speed in revolution per second (r.p.s) and given as $n = \frac{N}{60}$

For a given machine, the number of poles and the number of conductors per parallel path (Z/A) are constant. Hence, equation (5) can be written as:

$$E = K\varphi n$$

Where K is a constant and given as $K = \frac{PZ}{A}$

$E \propto \varphi n$ or $E = K_1 \varphi N$

$E \propto \varphi N$ or $E \propto \varphi \omega$

Thus, it is clear that the induced emf is directly proportional to the speed and flux per pole.

In DC Generator

wave wound generator, Numbers of parallel paths are only 2 = A

Induced emf for wave type of winding generator is $\frac{\phi PN}{60} X \frac{Z}{2} = \frac{\phi ZPN}{120} volts$

lap-wound generator, P = A

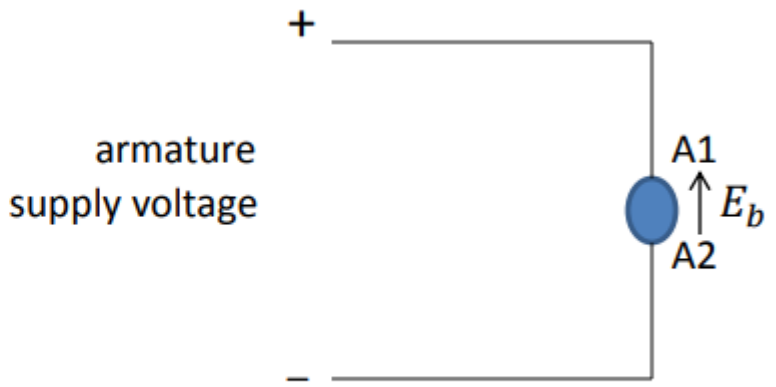
Induced emf for lap-wound generator is $E_g = \frac{\phi ZN}{60} X \frac{P}{A} volt$

Back emf

- When the armature winding of dc motor is start rotating in the magnetic flux produced by the field winding, it cuts the lines of magnetic flux and induces the emf in the armature winding.
- According to Lenz's law (The law that whenever there is an induced electromotive force (emf) in a conductor, it is always in such a direction that the current it would produce would oppose the change which causes the induced emf.), this induced emf acts in the opposite direction to the armature supply voltage. Hence this emf is called as back emfs.

$$E_b = \frac{N\emptyset Z}{60}\frac{P}{A} \text{ Volts}$$

N= speed in rpm
$\emptyset$= flux per pole
Z= no of conductors
P=no of pole pairs
A=area of cross section of conductor
E_b= back emf



Torque equation of DC Motor

Torque is defined as turning or twisting moment of a force about an axis. The wheel is rotating at a speed of N r.p.m. then angular speed of the wheel is ,

$$\omega = \frac{2\pi N}{60} \rightarrow \quad \text{rad / sec}$$

So work done in one revolution is,

W = F × Distance travelled in one revolution

$$W = F \times (2\pi R)$$

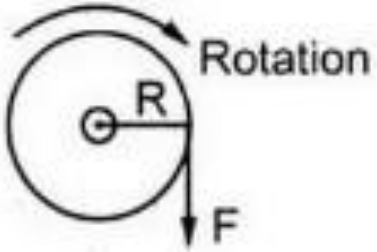


Fig. 4.7

And power developed = Workdone / time

$$P = \frac{(F \times 2\pi R)}{60} \times N$$

Torque , T=F *R

Power developed in armature = Ta *2πN/60……eq1

The voltage equation of motor, V = Eb + IaRa

Multiply with Ia on both sides, VIa = EbIa + Ia2 Ra ……..eq2

Comparing both the equations 1 and 2

$$E_b I_a = T_a \times \left(\frac{2\pi N}{60}\right)$$ …….eq3

But in motor, back emf given as

$$E_b = \phi N \left(\frac{ZP}{60A}\right)$$

Substituting the Eb value in the equation 3

$$\phi N \left(\frac{PZ}{60A}\right) \times I_a = T_a \times \left(\frac{2\pi N}{60}\right)$$

$$T_a = (\phi \times I_a) \times \left(\frac{ZP}{2\pi A}\right)$$ N-m

When the machine is operating as a motor, the torque is transferred to the shaft of the rotor and drives the mechanical load. The expression is the same for the generator and motor.

When the current-carrying current is placed in the magnetic field, a force is exerted or it which exerts turning moment or torque F x r. This torque is produced due to the electromagnetic effect, hence is called **Electromagnetic torque.**

The torque which is produced in the armature is not fully used at the shaft for doing the useful work. Some part of it gets lost due to mechanical losses. The torque which is used for doing useful work in known as the **shaft torque**

$$Output = T_{sh} \times 2\pi N$$

$$T_{sh} = \frac{Outputs\ in\ watts}{2\pi N}\ N-m\ in\ r.p.s$$

$$T_{sh} = \frac{Outputs\ in\ watts}{2\pi N/60}\ N-m\ in\ r.p.m$$

$$T_{sh} = \frac{60}{2\pi}\frac{Output}{N} = 9.55\frac{Output}{N}\ N-m\ in\ r.p.s$$

Lost torque: The difference between the armature torque and the shaft torque ($T_a - T_{sh}$) is known as the **lost torque** and is due to the formation of the torque.

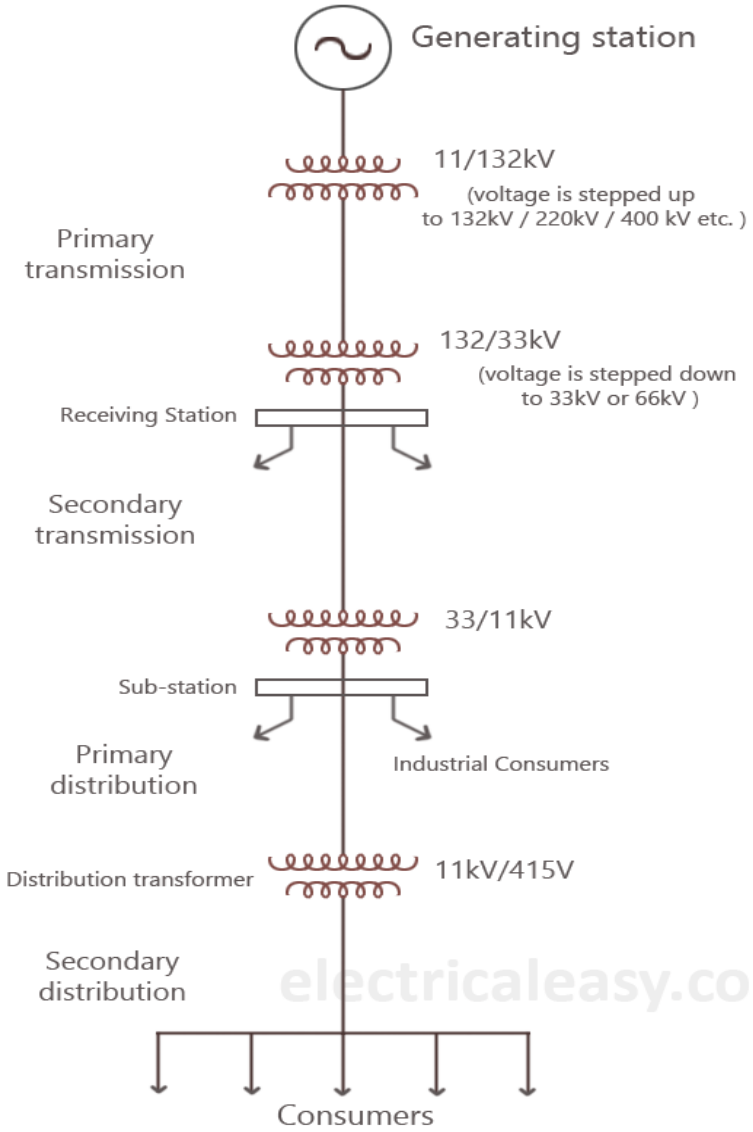
Applications of DC Motors

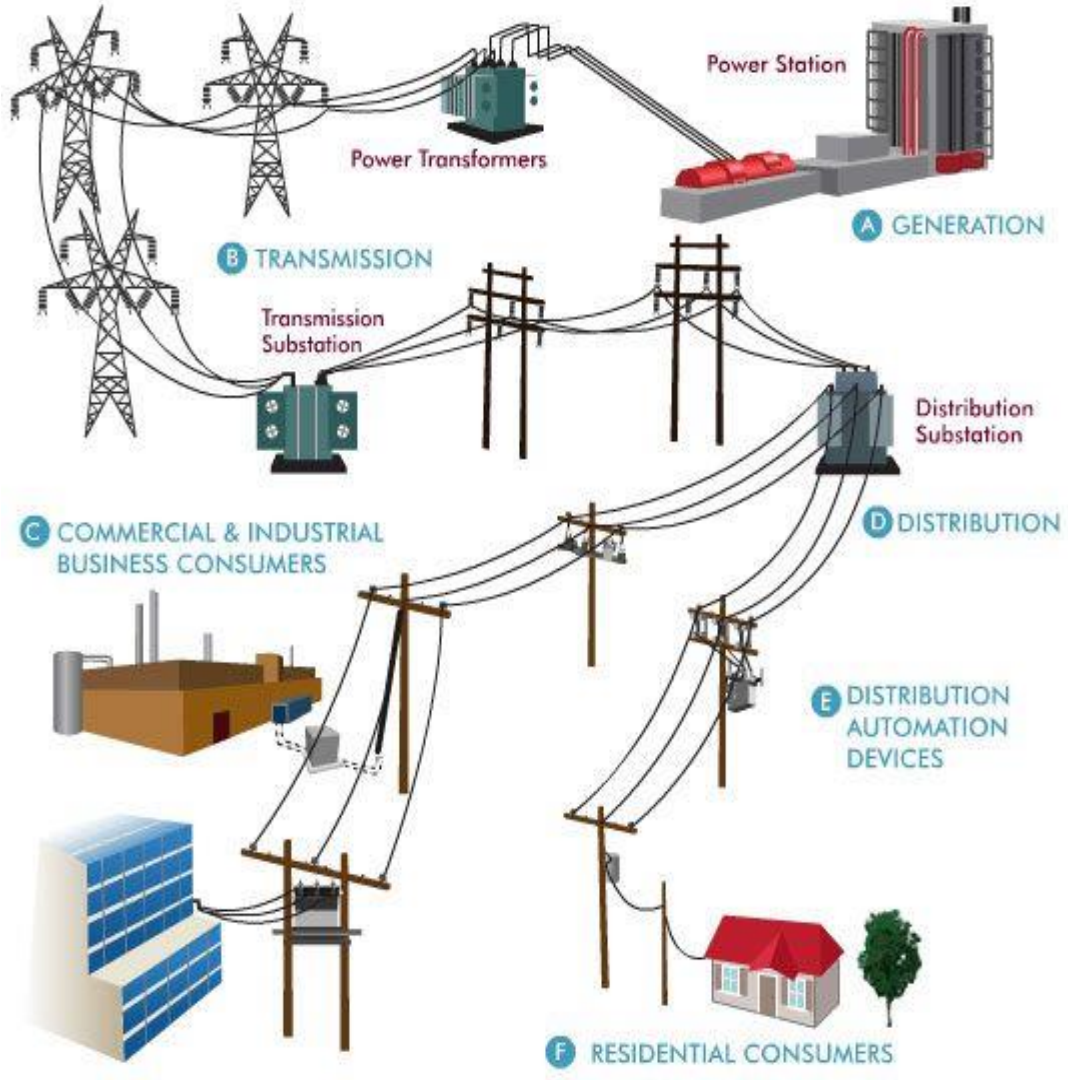
Type of Motor	Characteristics	Applications
Shunt	Speed is fairly constant and medium starting torque.	1. Blowers and fans 2. Centrifugal and reciprocating pumps 3. Lathe machines 4. Machine tools 5. Milling machines 6. Drilling machines
Series	High starting torque. No load condition is dangerous. Variable speed.	1. Cranes 2. Hoists, Elevators 3. Trolleys 4. Conveyors 5. Electric locomotives
Cumulative compound	High starting torque. No load condition is allowed.	1. Rolling mills 2. Punches 3. Shears 4. Heavy planers 5. Elevators
Differential compound	Speed increases as load increases.	Not suitable for any practical applications

TRANSFORMERS



Single line diagram of AC power transmission and distribution





Need of the transformer

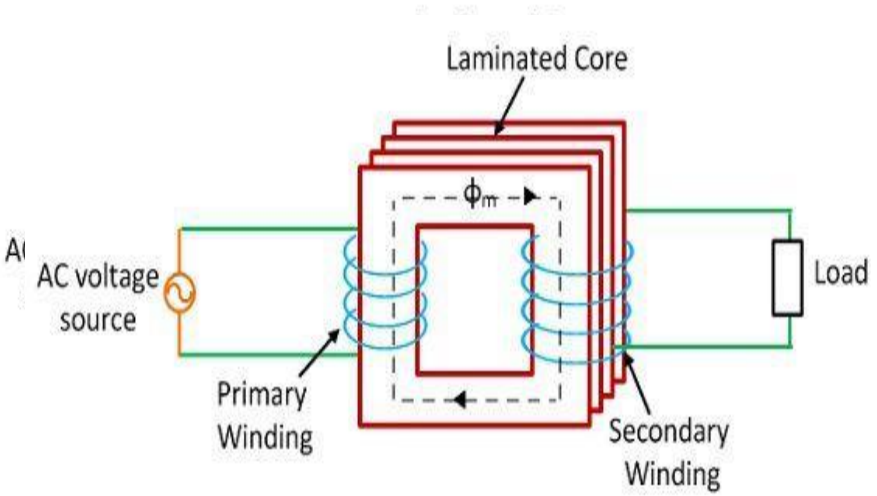
- In most cases, appliances are manufactured to work under some specific voltages. Transformers are used to adjust the voltages to a proper level.
- The transformers are the basic components for the transmission of the electricity.
- Transformer is used to increase the voltage at the power generating station (Step up) and used to decrease the voltage (Step down) for house hold purpose.
- By increasing the voltages the loss of the electricity in the transmission purpose is minimized.

What is Transformer?

- Transformer is a static device.
- It transfer electrical energy from one part of the electrical circuit to other part of circuit without changing the frequency and with the desired change in Voltage or current.
- The basic principle on which the transformer works is **Faraday's Law of Electromagnetic Induction** or **mutual induction between the two coils**.
- A transformer used to increase the voltage is called a **step up transformer.**
- A transformer used to decrease the voltage is called a **step down transformer.**

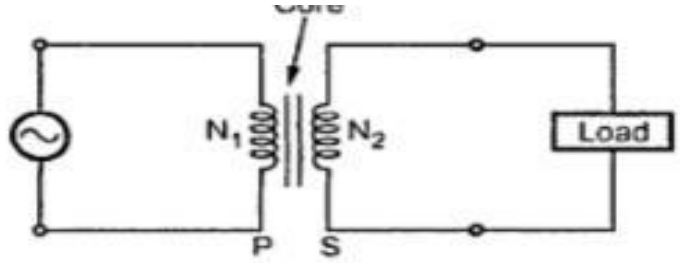
Working principle of transformer

- Transformer works on **Faraday's Law of Electromagnetic Induction** or **mutual induction between the two coils.**
- **Mutual induction principle:** when two coils are inductively coupled and if current in one coil is changed uniformly then an e.m.f. gets induced in the other coil. This e.m.f can drive a current, when a closed path is provided to it.
- The winding which is connected to the supply is knows as **Primary winding.**
- The winding which is connected to the load is knows as **Secondary winding**.
- It produce self induced emf in primary and mutual induced emf in secondary winding.

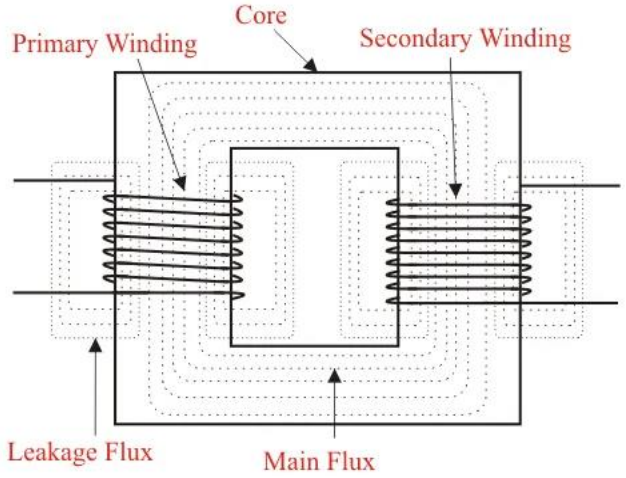


Electrical Transformer

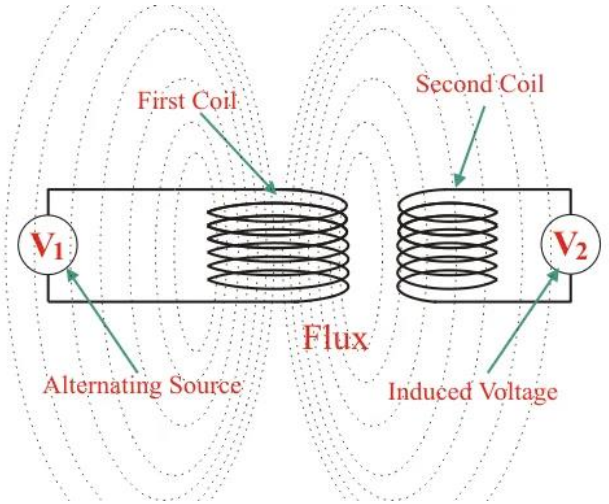
Circuit Globe



- When, primary winding is connected to a source of alternating voltage, alternating magnetic flux is produced around the winding. The core provides magnetic path for the flux, to get linked with the secondary winding. Most of the flux gets linked with the secondary winding which is called as 'useful flux' or main 'flux', and the flux which does not get linked with secondary winding is called as 'leakage flux'. As the flux produced is alternating (the direction of it is continuously changing), EMF gets induced in the secondary winding according to Faraday's law of electromagnetic induction. This emf is called 'mutually induced emf', and the frequency of mutually induced emf is same as that of supplied emf



- Physically, there is no electrical connection between the two windings, but they are magnetically connected.
- The induced emf in the primary and secondary windings depends upon the rate of change of flux linkage that is ($N \, d\phi/dt$).



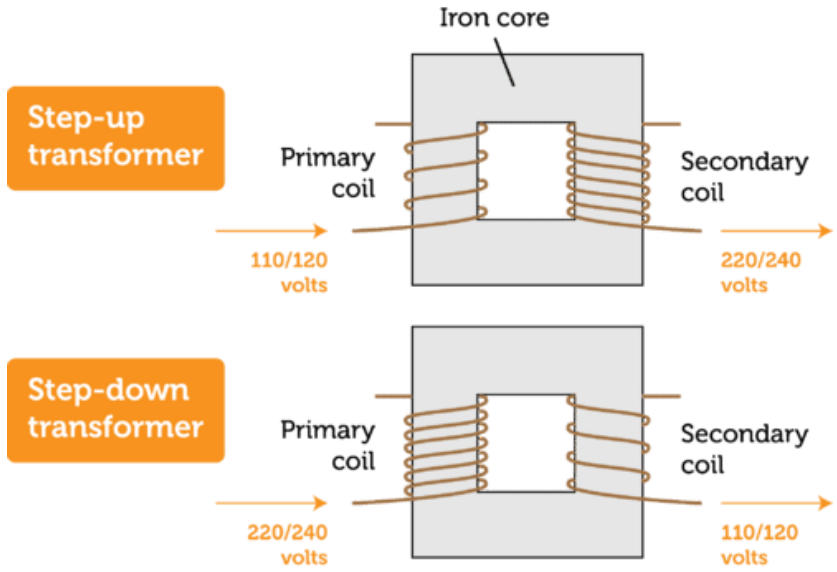
- $d\phi/dt$ is the change of flux and is same for both the primary and secondary windings.

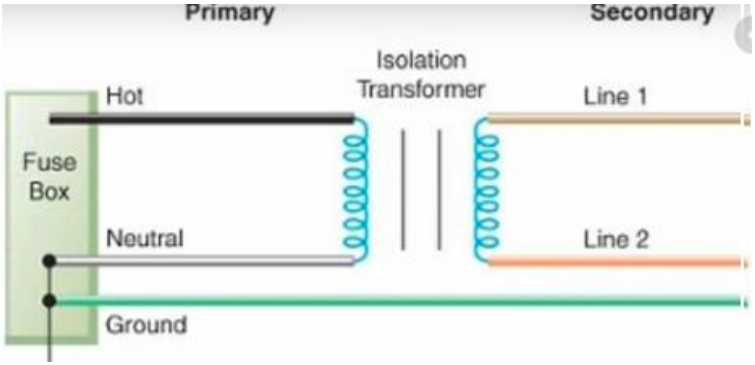
- The induced emf E_1 in the primary winding is proportional to the number of turns N_1 of the primary windings ($E_1 \propto N_1$).

- Similarly induced emf in the secondary winding is proportional to the number of turns on the secondary side. ($E_2 \propto N_2$).

Basis of winding

* **Step up Transformer:** The no of windings on Primary side(Np) is less than the no. of windings on the secondary side(Ns). Np < Ns
* **Step down Transformer:** The no. of winding on Primary side(Np) are more than the no. of windings on the secondary side(Ns). Np > Ns
* **Isolation Transformer:** The no. of winding on Primary side(Np) are equal to the no. of windings on the secondary side(Ns). Np = Ns





Transformer on DC supply

- The d.c. supply can not be used for the transformers.
- The transformer works on the principle of mutual induction, for which current in one coil must change uniformly. If d.c. supply is given, the current will not change due to constant supply and transformer will not work.
- Practically winding resistance is very small. For d.c., the inductive reactance X_L is zero as d.c. has no frequency. So total impedance of winding is very low for d.c. Thus winding will draw very high current if d.c. supply is given to it. This may cause the burning of windings due to extra heat generated and may cause permanent damage to the transformer.
  - There can be saturation of the core due to which transformer draws very large current from the supply when connected to d.c.
- Thus d.c. supply should not be connected to the transformers.

Transformer Classification

1.According to No. of phases

(a) Single phase transformer (b) Three phase transformer

2.According to Construction

(a) Core type transformer (b) Shell type transformer (c) Berry type transformer

3.Accoring to Function

(a)Power transformer----(i) Step-up transformer (ii) Step-down transformer

(b) Distribution transformer

(c) Instrument transformer ---(i) Current transformer (ii) Potential transformers

4. Special purpose transformers

(a)Scott connection (b) Vee connection

(c) Pulse transformers (d) Auto transformers (e) Welding transformers

5.According to Type of cooling

(a)Air cooled transformers (b)Oil cooled transformers

6.According to Location

(a)Indoor transformers (b)Outdoor transformers

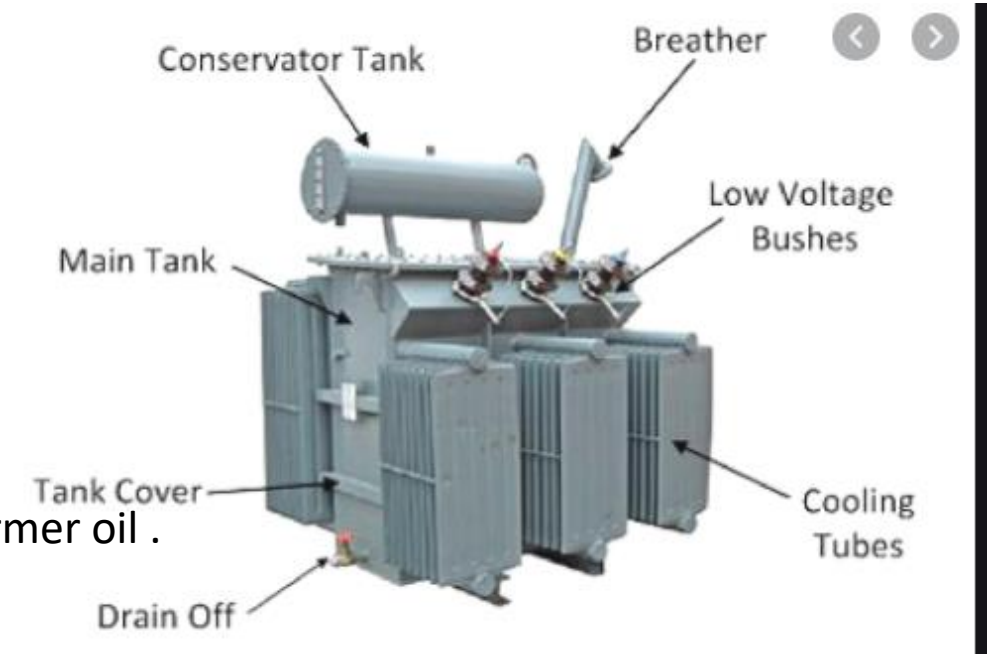
7.According to Frequency

(a)Low frequency(50Hz) (b)High frequency transformers

8.Accoring to Rating

(a) Low rating transformers (b) Medium rating transformers (c)High rating transformers

Parts of a transformer



1.Magnetic core: It provide a path of low reluctance for the magnetic flux.
Core is made up of laminated stampings to reduce eddy current losses.

2.Windings or coils: These are made up of copper and placed on core. Primary and secondary windings are there.

3.Transformer Tank: It is housed in proper container it has transformer oil .

4.Transformer Oil: It is also called Insulating oil.

(i) It carries away the heat produced in the core and windings
(ii) The oil should be absolutely free from alkalies, sulphur and specially from moisture.

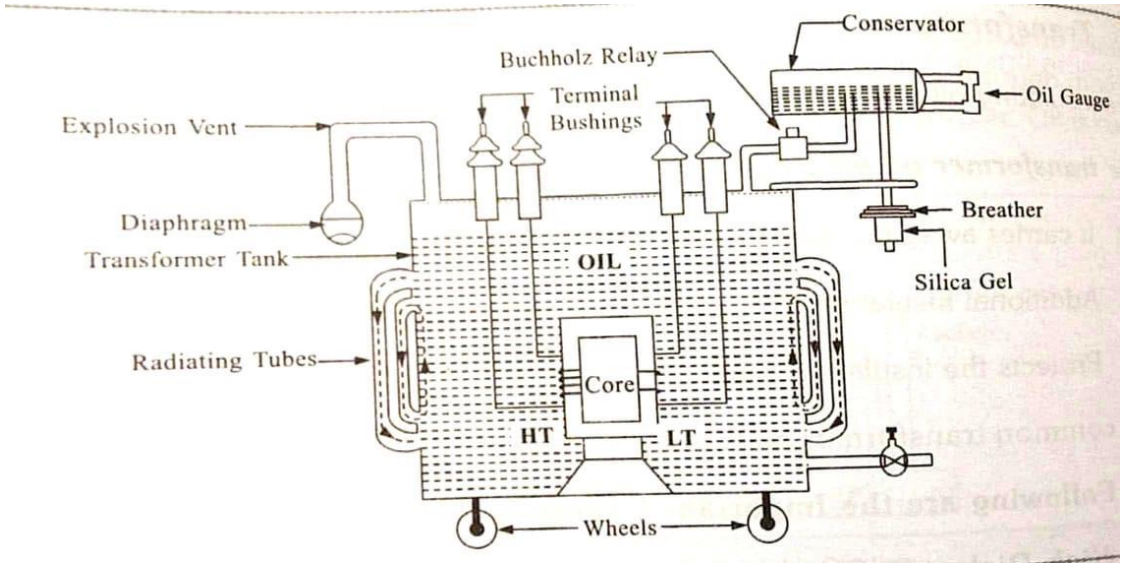
5.Conservator

(i)It maintain the oil level in the tank
(ii)It provides space for the expansion of oil, when temperature increases.
(iii)It reduces the rate of oxidation of oil because it ensures surface of the oil is less exposed to oil

6.Breather: It is to prevent entry of moisture in the air and allows dry air into the transformer.Moisture reduces the dielectric strength of the oil.

7.Terminal winding: Connections from the transformer windings are brought out by means of bushing mounted on transformer tank.

8.Explosion Vent : It provides instantaneous relieving of dangerous pressure and protects the transformer tank against explosion.



9.Oil level indicator: For indicating the level of the oil in the conservator an oil level indicator is fix to it. Oil lever should be maintained above predetermined minimum level.

10.Buchholz Relay: It is used for protection of oil filled transformers from incipient faults below oil lever

Basic parts of Transformer

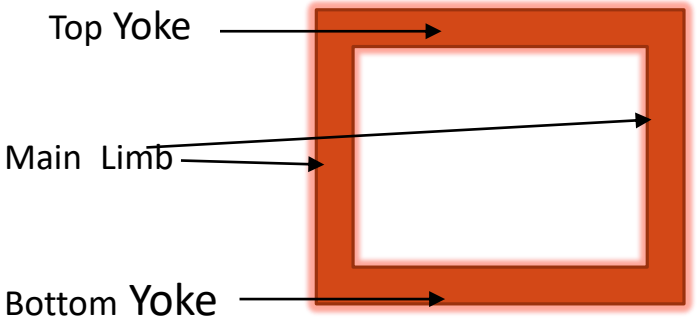
The two basic parts of a transformers are

1.Magnetic core 2.Windings or Coils

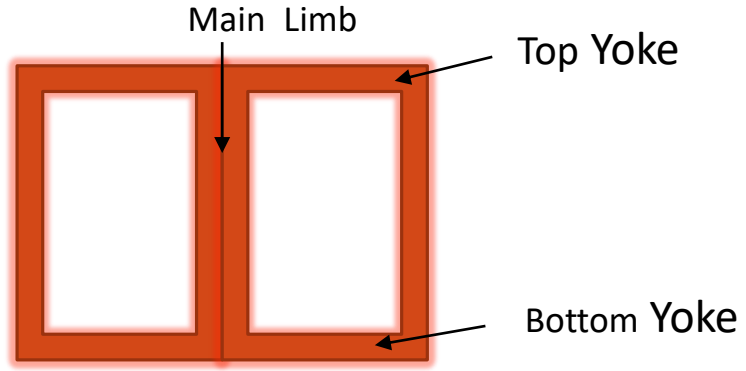
1.Magnetic core: It has two parts

(i) Limbs or Legs: The vertical portion on which windings or coils are wound is called as limbs.

(ii)Yokes: The top and bottom horizontal portions are called Yokes of the core.



1- Ø Two-limbed core(core type)



1- Ø Three-limbed core(shell type)

Core is made up of laminated stampings to reduce eddy current losses.

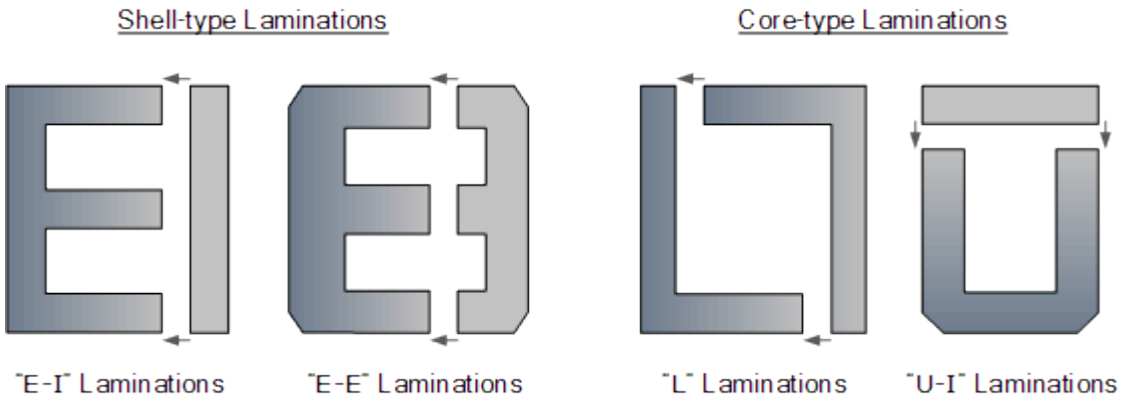
Eddy current loss: When there is a change in the magnetic field the emf circulates current within the body of the material.

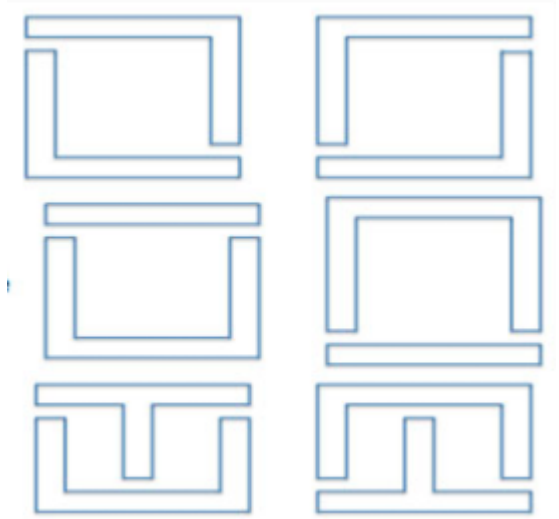
Generally, thin Silicon stell liminations(0.3-0.5mm thich are used)

Hysteresis loss: The energy which is wasted in the form of heat due to hysteresis is called Hysteresis Loss.

To reduce these losses we use Cold rolled grain oriented(CRGO) silicon steel laminations are used.

The laminated stampings are of different shapes like L,E,I,U,J,C etc.





2.Windings or Coils: The transformer has High voltage and Low voltage coils for each phase. The coils may be either

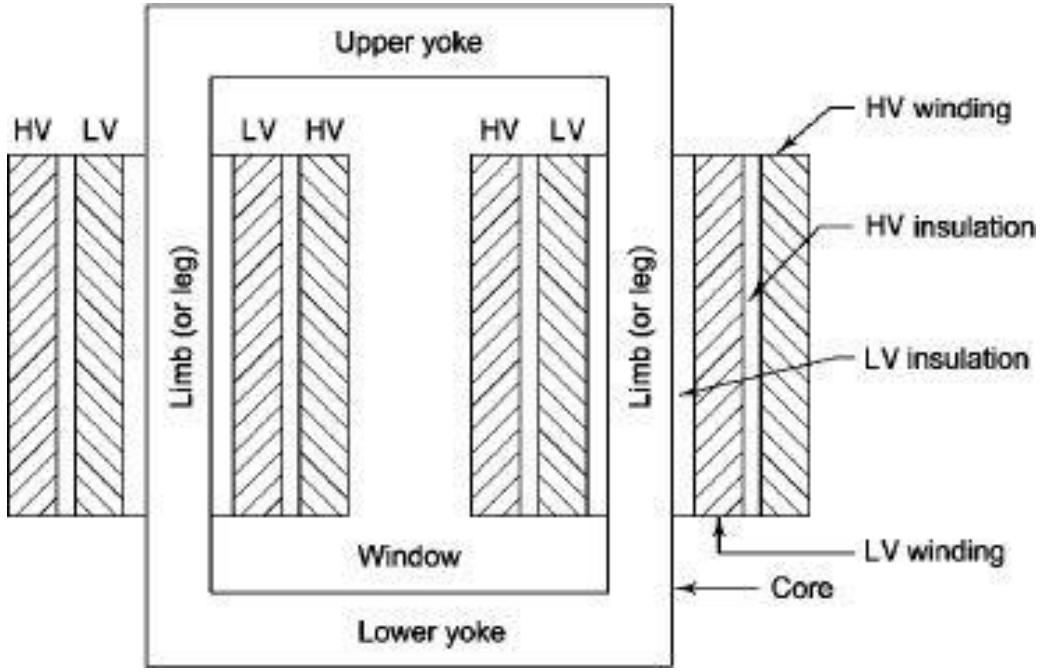
(i)cylindrical concentric (ii) Sandwiched type.

(i).Concentric coils:

- These are used in Core type transformers.
- Two windings are wound on two different Limbs due to this leakage flux increases and Hence performance of t/f affect badly.
- To reduce this flux, two windings are wound on the same limb in the transformer.

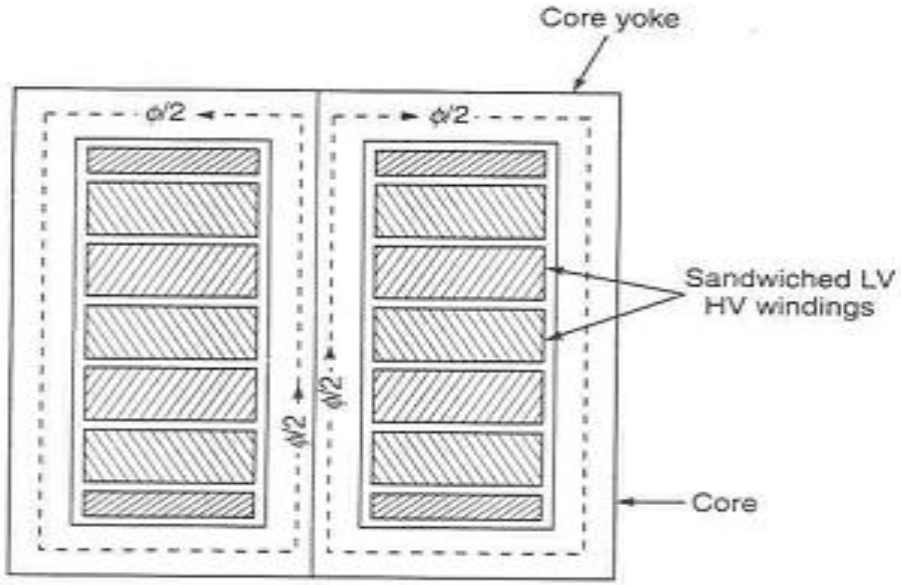
Here the position of LV and HV is important,

As LV winding is placed near the core for the Ease of insulating it from the core.
The HV winding is placed after it.

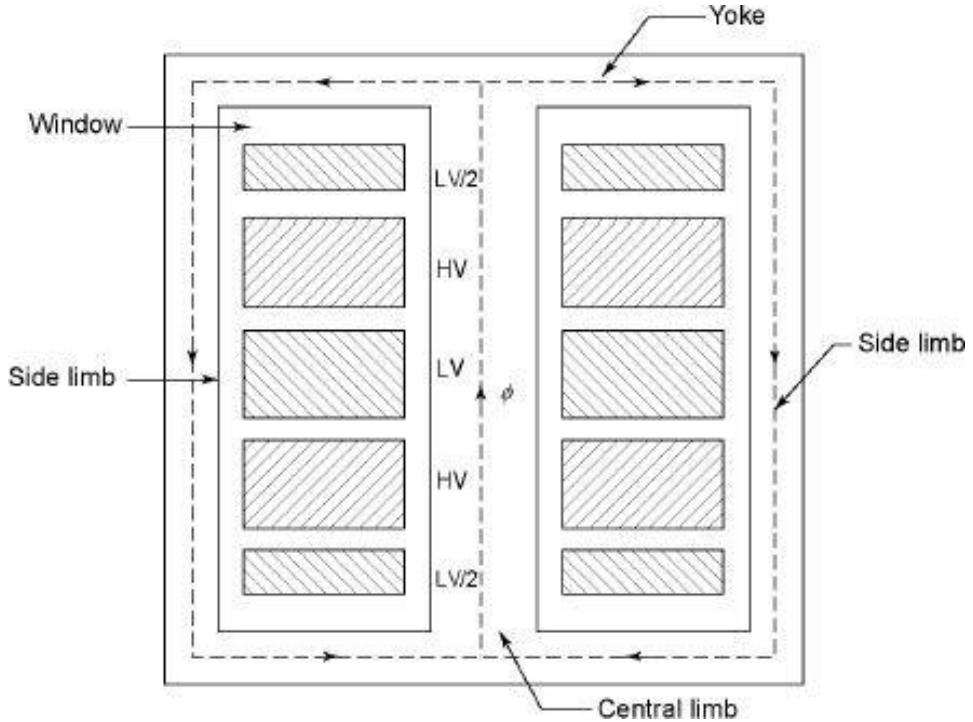


(ii) Sandwiched type

- These are used in Shell type.
- The leakage reactance of the windings can be easily controlled by employing this type winding.
- The nearer the HV and LV coils, the less is the Leakage flux.
- The HV sections lie between two consecutive low voltage sections.
- The two end sections are LV sections and contain half the turns of other LV sections.
- The reactance values can be decreased by increasing the number of sub divisions.



Shell Type Transformer



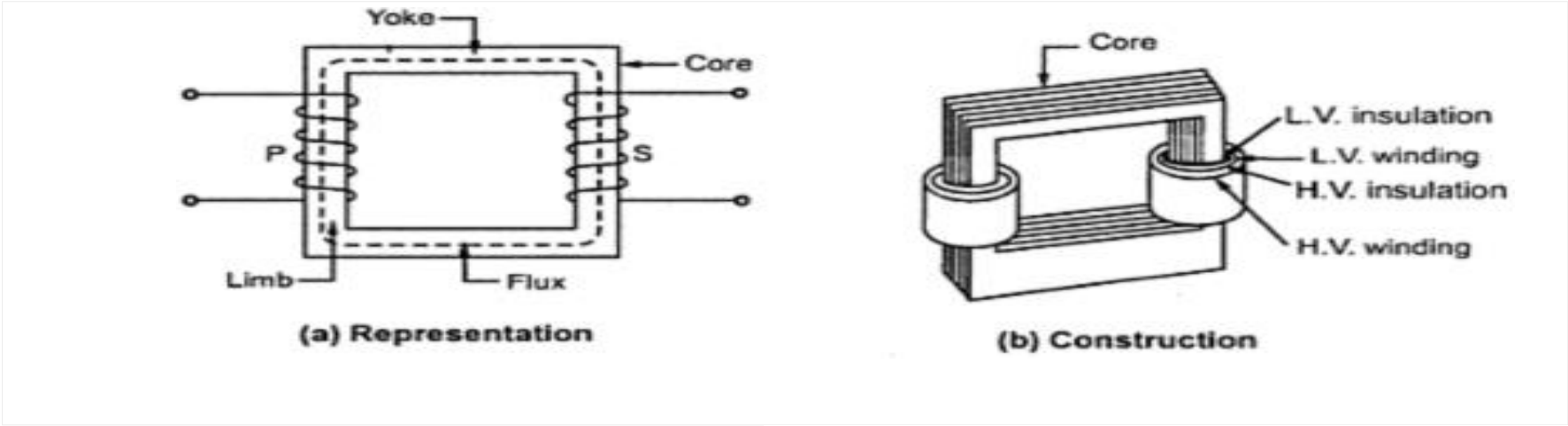
Construction of Transformer

The various constructions used for the single phase transformers are,

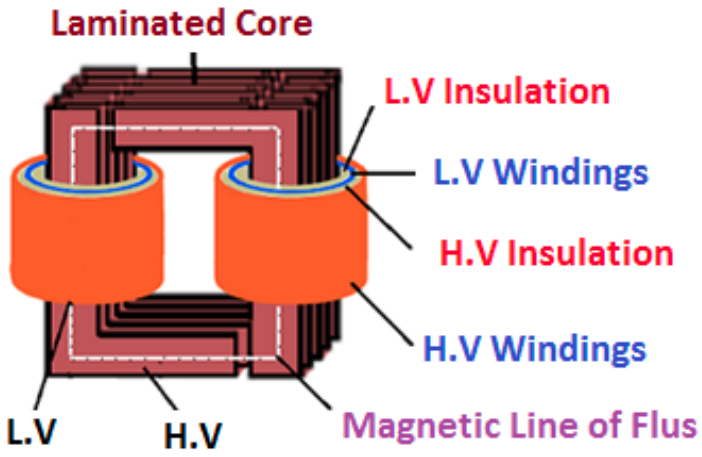
1. Core type 2. shell type and 3. Berry type

1.Core type transformer:

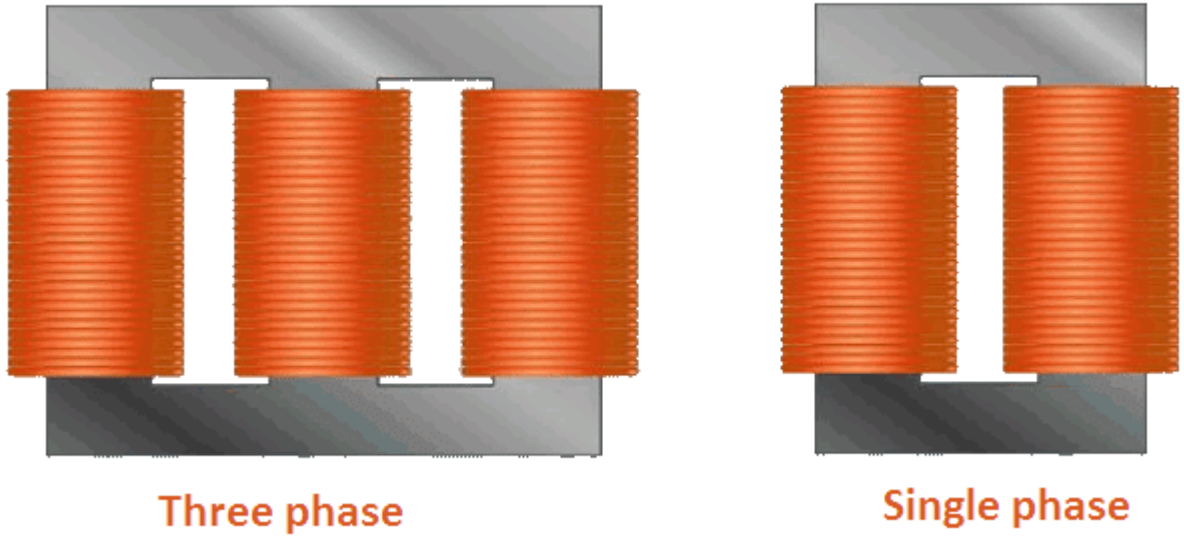
- It has a single magnetic circuit. The core rectangular having two limbs.
- **The winding encircles the core**. The coils used are of cylindrical type.
- The coils are wound in helical layers with different layers insulated from each other by paper or mica.
- The LV winding is placed near the core for the Ease of insulating it from the core. The HV winding is placed after it.



Core type transformer



Core Type Transformer

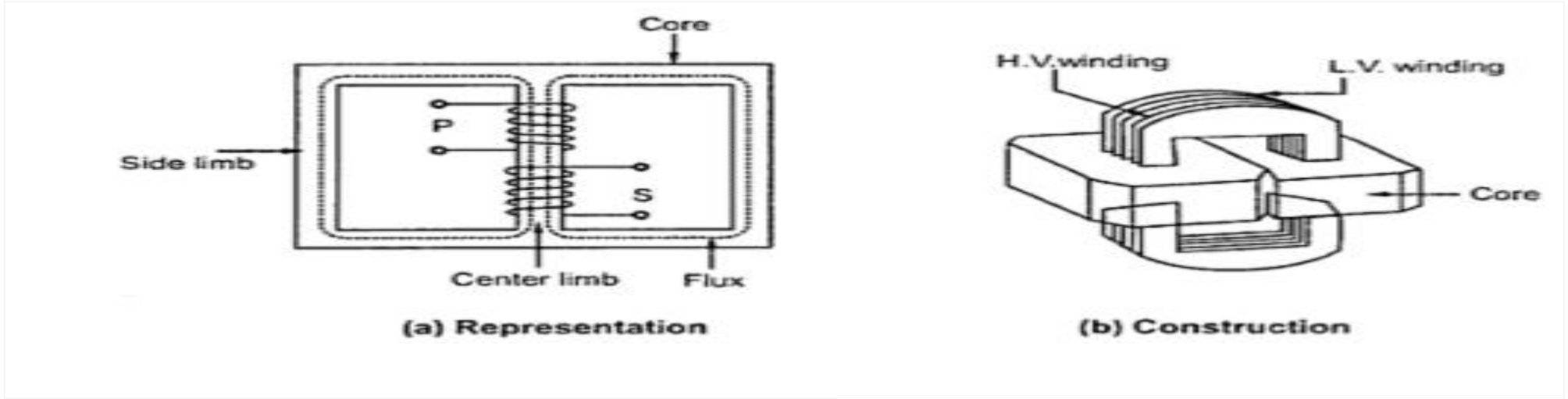


Three phase

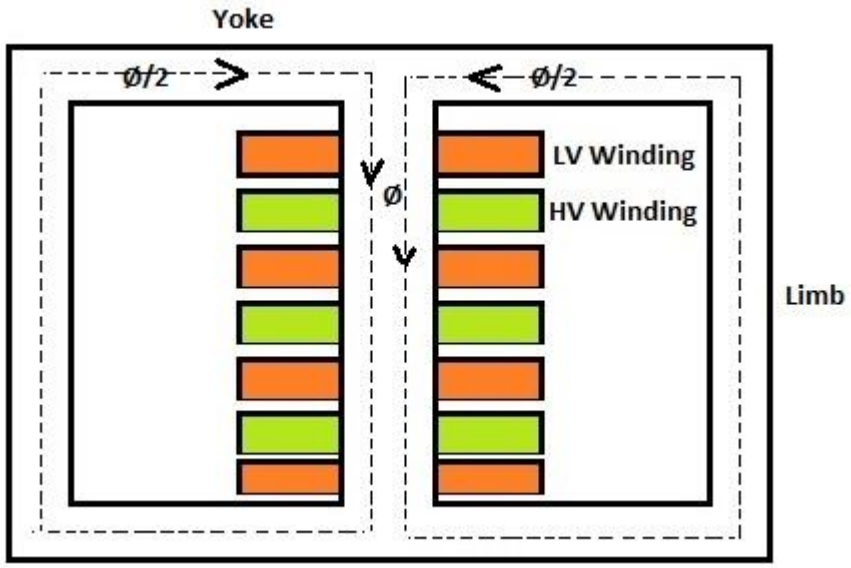
Single phase

2.Shell Type Transformer

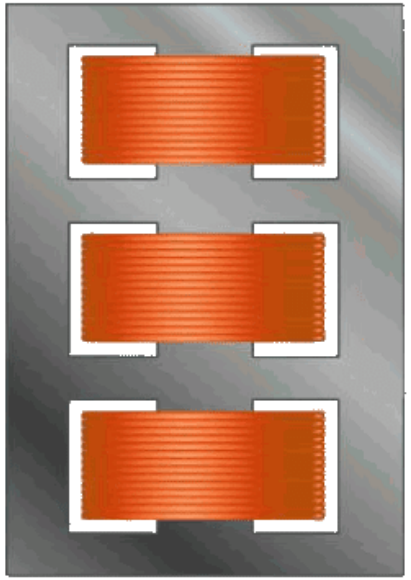
- It has a double magnetic circuit. The core has three limbs. Both the windings are placed on the central limb.
- The core encircles most part of the windings.
- The coils used are generally multilayer disc type or sandwich coils.
- Generally for very high voltage transformers, the shell type construction is preferred.
- In this type of winding, natural cooling does not exist. For removing any winding for maintenance, large number of laminations are required to be removed.



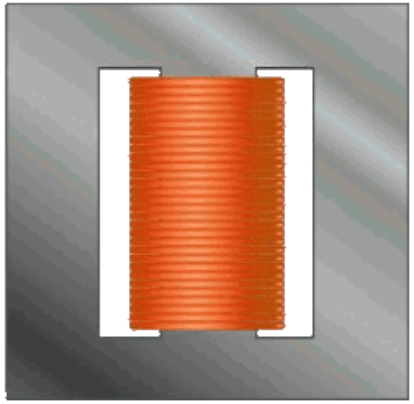
Shell type transformer



Shell Type Transformer



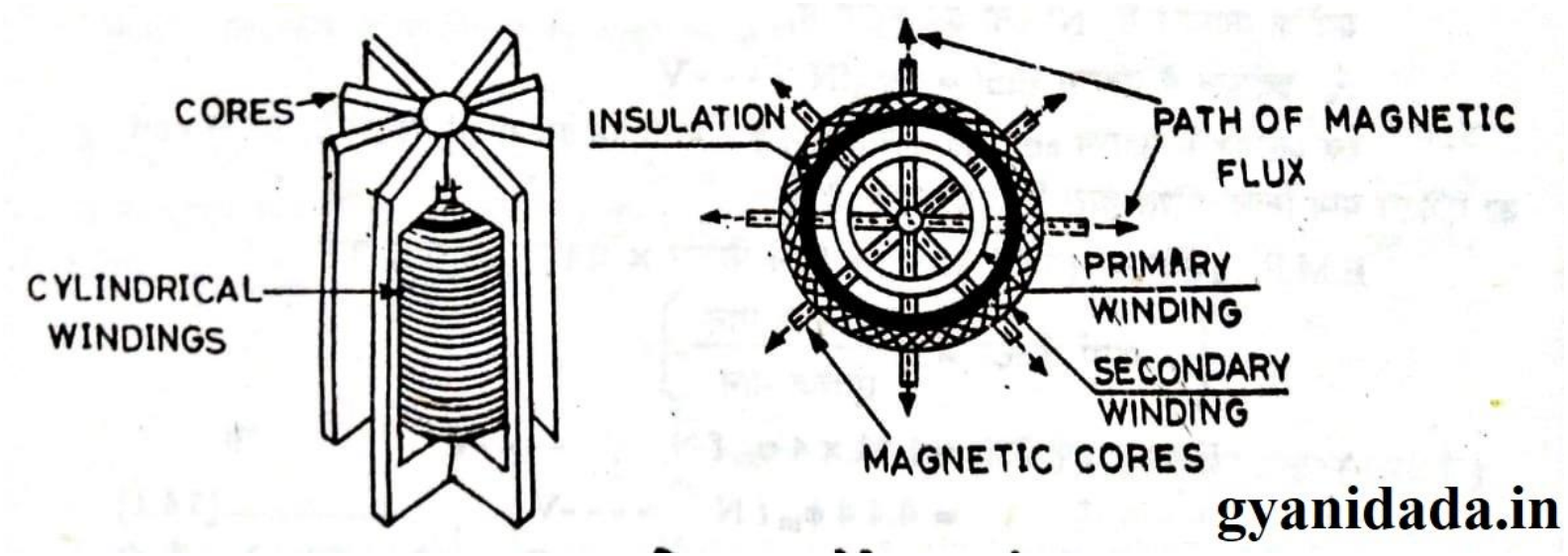
Three phse

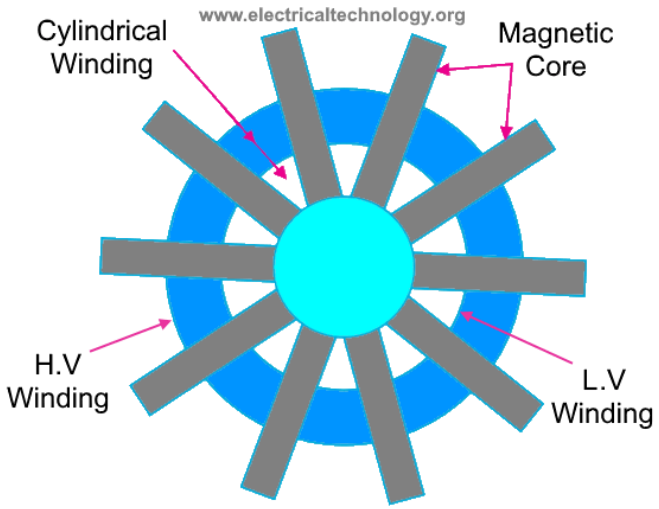


Single phase

3.Berry Type Transformer

- This has **distributed magnetic circuit.**
- The number of independent magnetic circuits are more than 2.
- Its core construction is like spokes of a wheel.
- This type of transformer has magnetic cores in the shape of rectangular frames.
- The transformers are generally placed in tightly fitted sheet metal tanks.
- The transformer is having multi magnetic paths.





Berry Type Transformer

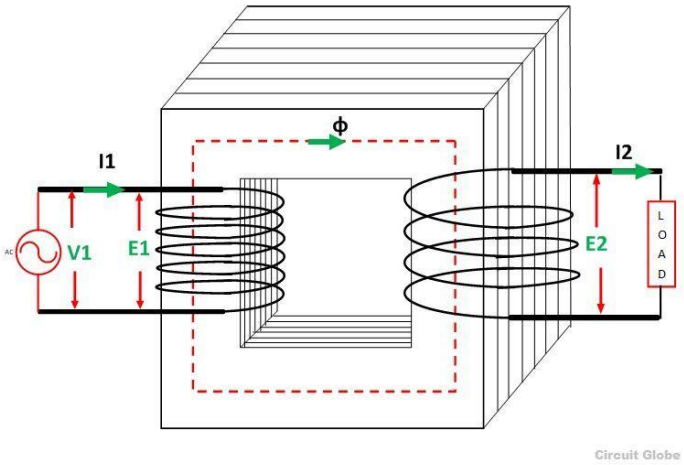
Differences between Core type and Shell type transformers

Core type	Shell type
1.The winding encircles the core.	The core encircles most part of the winding.
2.The cylindrical type of coils are used.	Sandwich type or Multilayer disc type coils are used.
3.As windings are distributed, the natural cooling is more effective.	As windings are surrounded by the core the natural cooling does not exist.
4.The coils can be easily removed from maintenance point of view.	For removing any winding for the maintenance, large number of laminations are required to be removed. This is difficult.
5.The construction is preferred for low voltage transformers	The construction is used for very high voltage transformers.
6.It has a single magnetic circuit	It has a double magnetic circuit.
7.In a single phase type, the core has two limbs.	In a single phase type, the core has three limbs.

E.M.F. Equation of a Transformer

When the primary winding is excited by an alternating voltage V_1, it circulates alternating current, producing an alternating flux Φ.

The primary winding has N_1number of turns. The alternating flux Φ linking with the primary winding itself induces an e.m.f in it denoted as E_1.

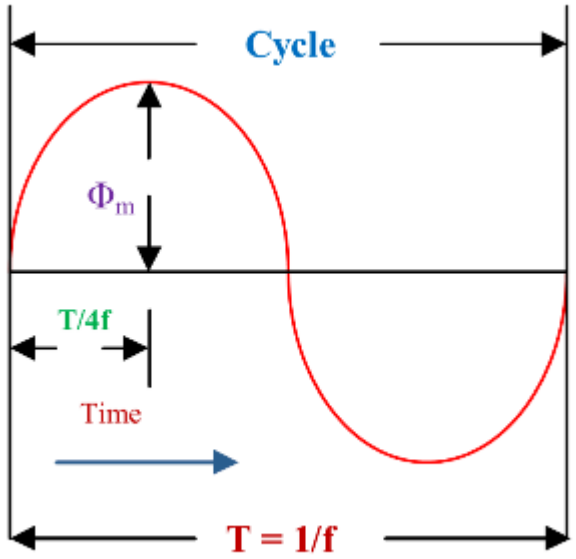


The flux links with secondary winding through the common magnetic core. It produces induced e.m.f. E_2 in the secondary winding.

This is mutually induced e.m.f. Let us derive the equations for E_1 and E_2.

The primary winding is excited by purely sinusoidal alternating voltage. Hence the flux produced is also sinusoidal in nature having maximum value of Φ_{max} show in the waveform.

The various quantities which affect the magnitude of the induced e.m.f. are :



Φ = Flux
Φm = Maximum value of flux
N_1 = Number of primary winding turns
N_2 = Number of secondary winding turns
f = Frequency of the supply voltage
E_1 = R.M.S. value of the primary induced e.m.f.
E_2 = R.M.S. value of the secondary induced e.m.f.

From Faraday's law of electromagnetic induction the voltage e.m.f. induced in each turn is proportional to the average rate of change of flux.

$\therefore$ average e.m.f. per turn = average rate of change of flux
$\therefore$ average e.m.f. per turn = $d\Phi/dt$
Now $d\Phi/dt$ = Change in flux/Time required for change in flux

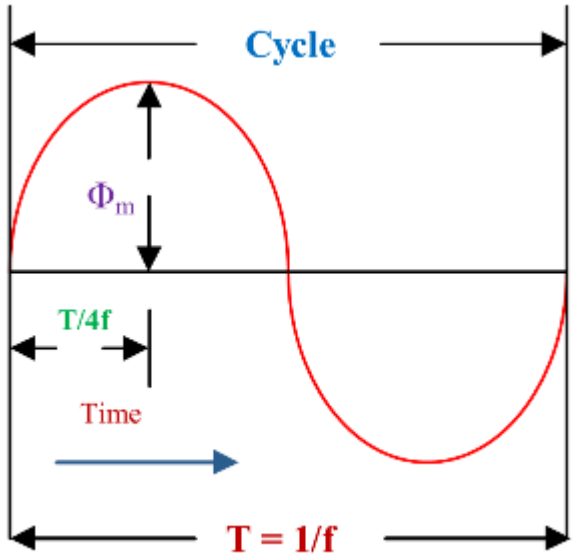
Consider the 1/4 th cycle of the flux as shown in the Fig. Complete cycle gets completed in 1/f seconds. In 1/4 th time period, the change in flux is from 0 to Φm.

∴ $d\Phi/dt = (\Phi m - 0)/(1/4f)$

as dt for 1/4 th time period is 1/4f seconds

$= 4 f \Phi m$ Wb/sec

∴ Average e.m.f. per turn = 4 f Φm volts



As is sinusoidal, the induced e.m.f. in each turn of both the windings is also sinusoidal in nature. For sinusoidal quantity,

From factor = R.M.S. value/Average value = 1.11

. = R.M.S. value of induced e.m.f.per turn

= 1.11 x 4 f Φm

= 4.44 f Φm

There are number of primary turns hence the R.M.S value of induced e.m.f. of primary denoted as is E_1,

$E_1 = N_1 \times 4.44\ f\ \Phi m$ volts

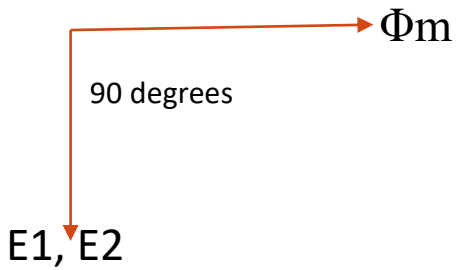
While as there are number of secondary turns the R.M.S values of induced e.m.f. of secondary denoted is E_2 is,

$E_2 = N_2 \times 4.44\ f\ \Phi m$ volts

The expression of E1 and E2 are called e.m.f. equation of a transformer. Thus e.m.f. equations are,

$E_1 = 4.44\ f\ \Phi m\ N_1$ volts .....(1)

$E_2 = 4.44\ f\ \Phi m\ N_2$ volts .....(2)



Phasor diagram

Note: If Bm is the maximum flux density in wb/m2 and A is the are of cross section in square meter's, then

$\Phi m = Bm * A$

Voltage and current Transformation Ratio

E1 = 4.44 f Φm N1 volts.....(1)

E2 = 4.44 f Φm N2volts.....(2)

$$\text{Turn ratio} = \frac{N_1}{N_2}$$

Taking the ration of these two equations.
The transformation ratio is defined as the ratio of the secondary voltage to the primary voltage.
It is denoted by K..

Transformation ratio, $K = \frac{N2}{N1} = \frac{V2}{V1} = \frac{E2}{E1} = \frac{I1}{I2}$

As (E_2 α N_2 and E_1 α N_1)

If $N_2 > N_1$, **E2>E1, K>1**, the transformer is called **Step-up transformer , Here $I_1 > I_2$**
If $N_2 < N_1$,**E2<E1, K<1**,the transformer is called **Step down transformer , Here $I_1 < I_2$**
If N2=N1,E2=E1, K=1, the transformer is called **isolation transformer , Here $I_1 = I_2$**

Problem1: A 400/230 V , 50Hz, single phase transformer has 250 turns on the low voltage side. If the maximum flux density in the core is 1.4 vwb/m2 .Calculate

(i) The cross section are of the core

(ii)The primary number of turns ratio

Problem2 :The emf per turn of 260/117 v, 1 kvA single phase power transformer is approximately 13 volts calculate

(i) The number of primary and secondary turns

(ii) The net cross sectional area of the core, for a maximum flux density is 1.4 tesla

Ideal Transformer

- The transformer which is free from all types of losses is known as an **ideal transformer**. The windings of the transformer are purely inductive.
- It is an imaginary transformer that has no core loss, no ohmic resistance, and no leakage flux.

The ideal transformer has the following **important characteristics**.

1. It has no core/iron losses(i.e iron losses= Eddy current loss +Hysteresis loss = 0)

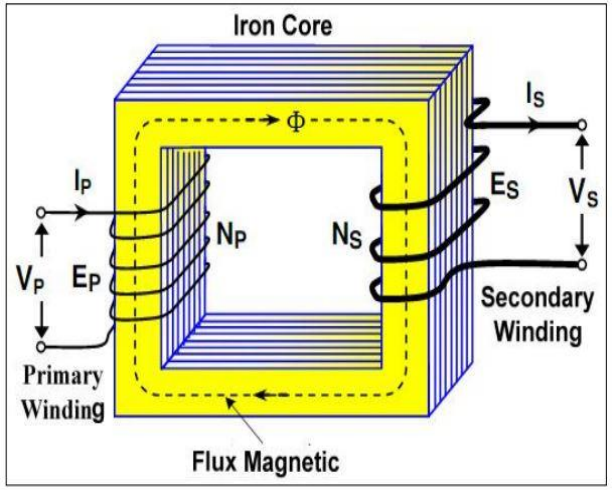
2.The resistance of their primary and secondary winding becomes zero. So there is no resistive voltage drop. (R=0)

3. The leakage flux of the transformer becomes zero, i.e. the whole of the flux induces in the core of the transformer links with their primary and secondary winding.(øleakage= 0)

4.Permeability of core is so high that means negligible current required to establish the flux in it.

5.Infinite permeability of the core, if the high permeability then less mmf required for flux established.

6.. It has 100 percent efficiency, i.e., power output= power input(100% Efficiency)



Ideal transformer: Ratios

- The primary applied voltage V1 and induced emf E1 are equal in ideal transformer, in the same way Seconday voltage V2and induced emf E2 also equal.

$$\frac{E_2}{E_1} = \frac{V_2}{V_1} = K$$

- There are no losses in ideal transformer so we conclude V_1I_1 = input VA and V_2I_2 = output VA

$$V_1I_1 = V_2I_2$$

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = K$$

Voltage ampere rating:

Thus the transformer rating is specified as the product of voltage and current called VA rating.
On both sides, primary and secondary VA rating remains same. This rating is generally expresses in KVA (kilo volt amperes rating).
Now V1 /V2 = I2 /I1 = K
∴ V1 I1 = V2 I2

$$\text{kVA rating of a transformer} = \frac{V_1 I_1}{1000} = \frac{V_2 I_2}{1000}$$

If V1 and V2 are the terminal voltages of primary and secondary then from specified KVA rating we can decide full load currents of primary and secondary, I1and I2. This is the safe maximum current limit which may carry, keeping temperature rise below its limiting value.

$$I_1 \text{ full load} = \frac{\text{kVA rating} \times 1000}{V_1} \quad \text{... (1000 to convert kVA to VA)}$$

$$I_2 \text{ full load} = \frac{\text{kVA rating} \times 1000}{V_2}$$

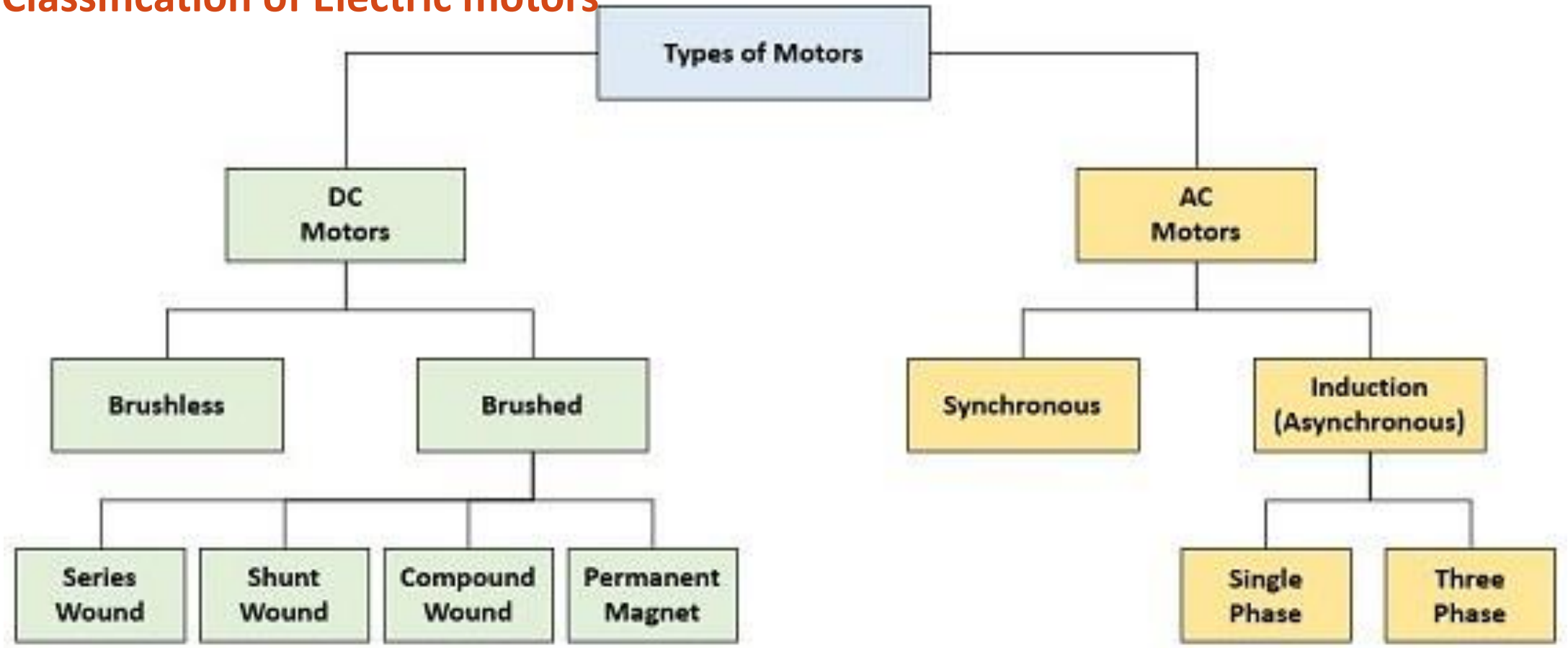
The full load primary and secondary currents indicate the safe maximum values of currents which transformer windings can carry.

Problem1: For a single phase transformer having primary and secondary turns of 440 and 880 respectively, determine the transformer KVA rating if half load secondary current is 7.5 A and maximum value of core flux is 2.25 Wb.

Problem2: A 100kvA, 11 kv/1400V,50Hz, 1-phase transformer has 110 turns of the secondary. Calculate the approximate values of the primary and secondary full load currents, the maximum value of the flux in the core and no. of primary turns.

3-Phase Induction Motor

Classification of Electric motors

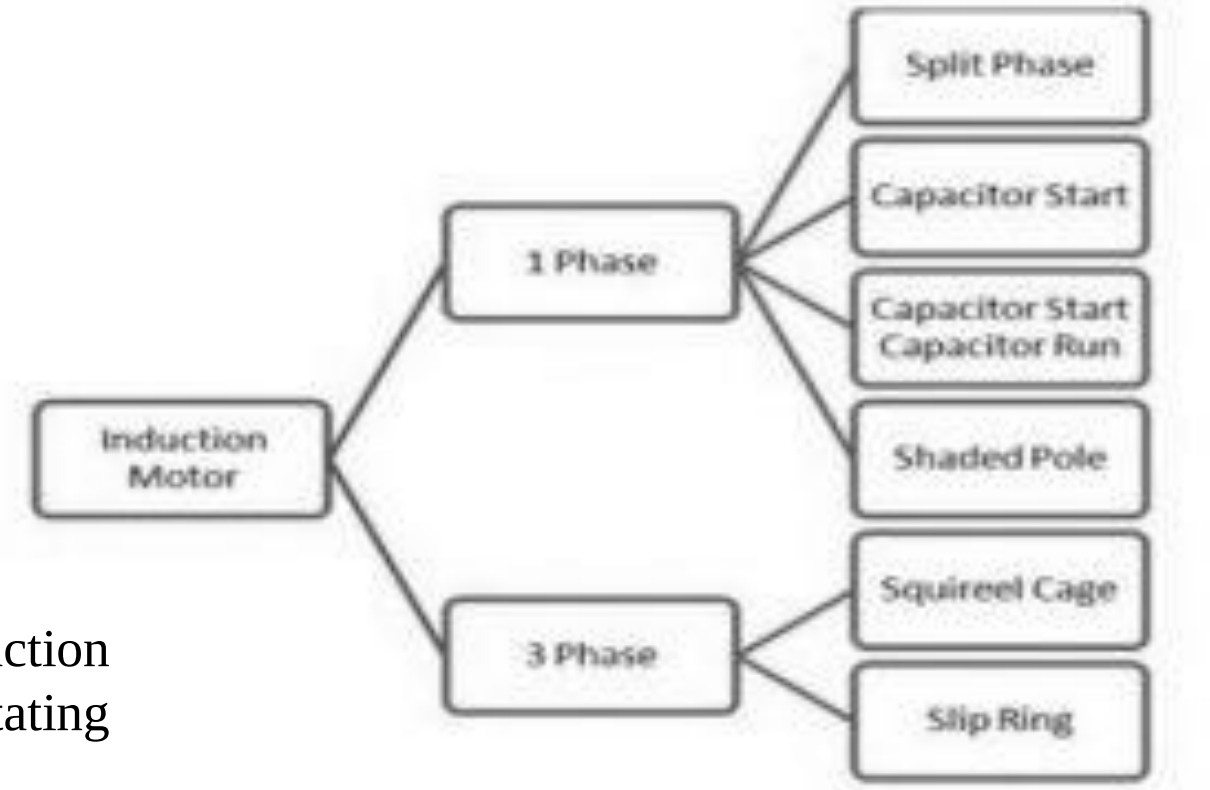


Three phase Induction Motors

The three phase induction motors are widely used for various industrial application.
The important features of three phase induction motors are

- Self starting
- Higher power factor,
- Good speed regulation
- Robust construction.
- Cost is low and reliable

The working of three phase induction motors is based on the principle of rotating magnetic field.



Rotating Magnetic field(R.M.F.)

When a 3-phase winding is energized from a 3-phase supply, a rotating magnetic field is produced. This field is such that its poles do no remain in a fixed position on the stator but go on shifting their positions around the stator. For this reason, it is called a rotating field.

The magnitude of this rotating field is constant and is equal to **1.5** $\emptyset_m$ where $\emptyset_m$ is the maximum flux due to any phase.

The rotating magnetic field can be defined as the field or flux having constant amplitude but whose axis is continuously rotating in a plane with a certain speed.

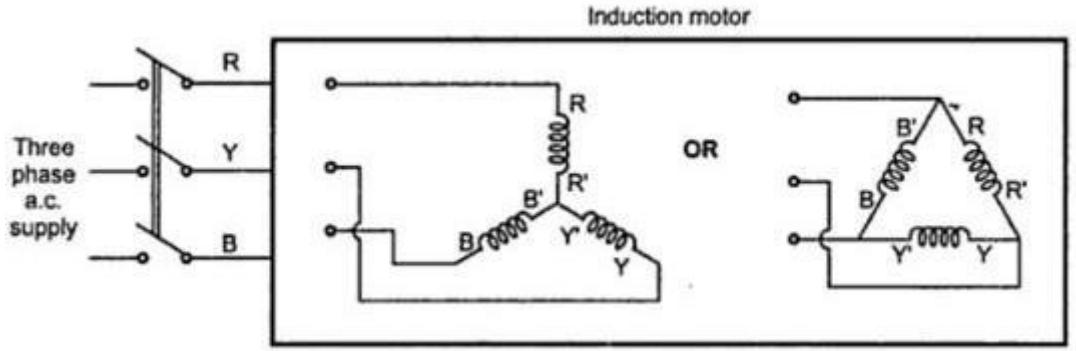
So if the arrangement is made to rotate a permanent magnet, then the resulting field is a rotating magnetic field.

Production of R.M.F.

- A three-phase induction motor consists of three phases winding as its stationary part called stator. The three-phase stator winding is connected in star or delta.

- The three-phase windings are displaced from each other by 120°. The windings are supplied by a balanced three phase ac supply.

- The three-phase currents flow simultaneously through the windings and are displaced from each other by 120° electrical. Each alternating phase current produces its own flux which is sinusoidal.
- So all three fluxes are sinusoidal and are separated from each other by 120°



Star and delta connection of 3-phase winding

If the phase sequence of the windings is R-Y-B, then mathematical equations for the instantaneous values of the three fluxes Φ_R, Φ_Y, Φ_B can be written as,

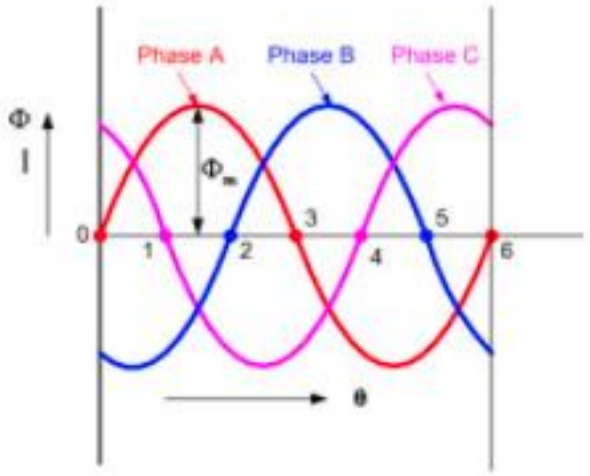
$$\Phi_R = \Phi m \sin(\omega t) = \Phi m \sin \theta$$

$$\Phi_Y = \sin (\omega t - 120^o) = \Phi m \sin (\theta - 120^o)$$

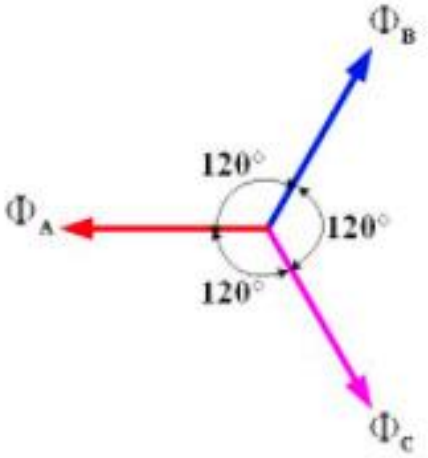
$$\Phi_B = \Phi m \sin (\omega t - 240^o) = \Phi m \sin (\theta - 240^o)$$

The effective or total flux (ϕ_T) in the air gap is equal to the phasor sum of the three components of fluxes ϕ_R, ϕ_Y and, ϕ_B.

Therefore, $\phi_T = \phi_R + \phi_Y + \phi_B$



Flux wave form



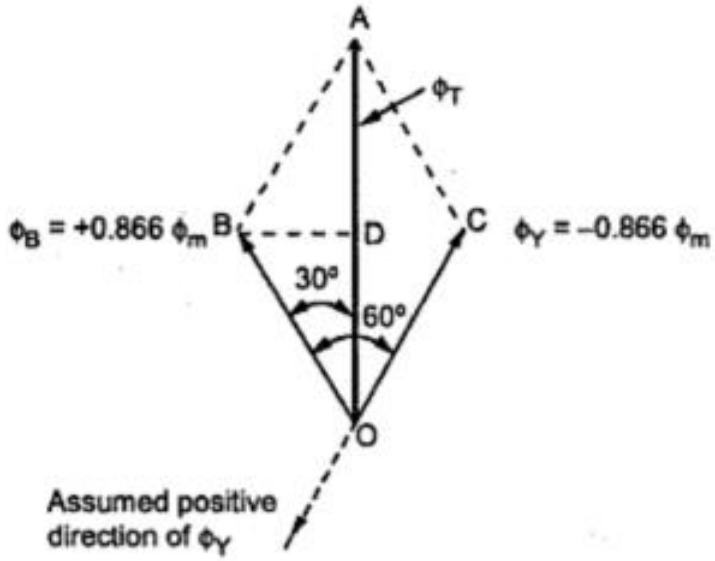
Positive direction of flux

Case 1: $\omega t = 0^o$ or $\theta = 0^o$

$\Phi_R = \Phi m \sin 0^o = 0$

$\Phi_Y = \Phi m \sin(-120^o) = -0.866\ \Phi m = -\sqrt{3}/2\ \Phi m$

$\Phi_B = \Phi m \sin(-240^o) = +0.866\ \Phi m = \sqrt{3}/2\ \Phi m$



Vector diagram of $\theta = 0^o$

BD is drawn perpendicular from B on Φ_T. It bisects Φ_T.

$\therefore \quad OD = DA = \Phi_T/2$

In triangle $\quad \angle OBD = 30^o$

$\therefore \quad \cos 30^o = OD/OB = (\Phi_T/2)/(0.866\ \Phi_m)$

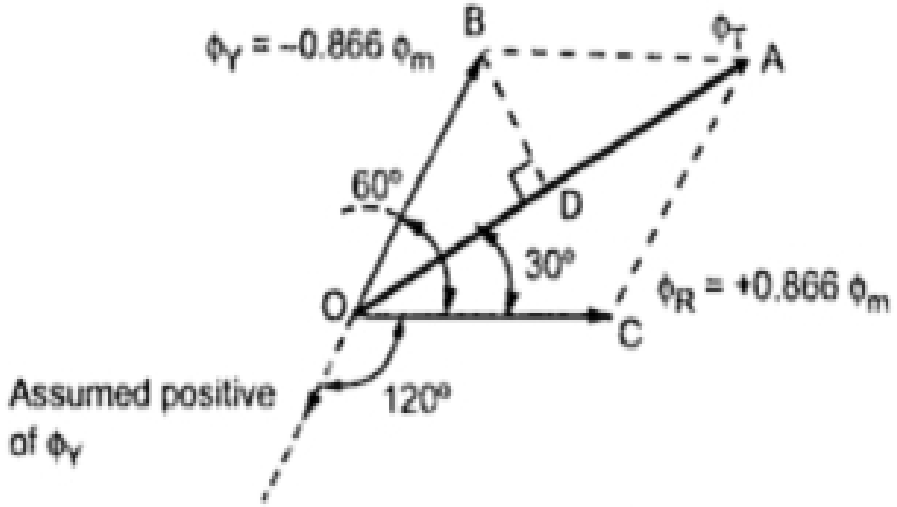
$\therefore \quad \Phi_T = 2 \times 0.866\ \Phi_m \times \cos 30^o$

$= 1.5\ \Phi_m$

So magnitude of Φ_T is $1.5\ \Phi_m$ and its position is vertically upwards at $\theta = 0^o$.

Case 2: ωt = 60° or θ = 60°

$\Phi_R = \Phi m \sin 60^o = +0.866\ \Phi m$
$\Phi_Y = \Phi m \sin (-60^o) = -0866\ \Phi m$
$\Phi_B = \Phi m \sin (-180^o) = 0 \Phi B = \Phi m \sin (-180^o) = 0$



Vector diagram of θ = 60°

Doing the same construction, drawing perpendicular from B on at D we get the same result as,

$\Phi_T = 1.5\ \Phi m$

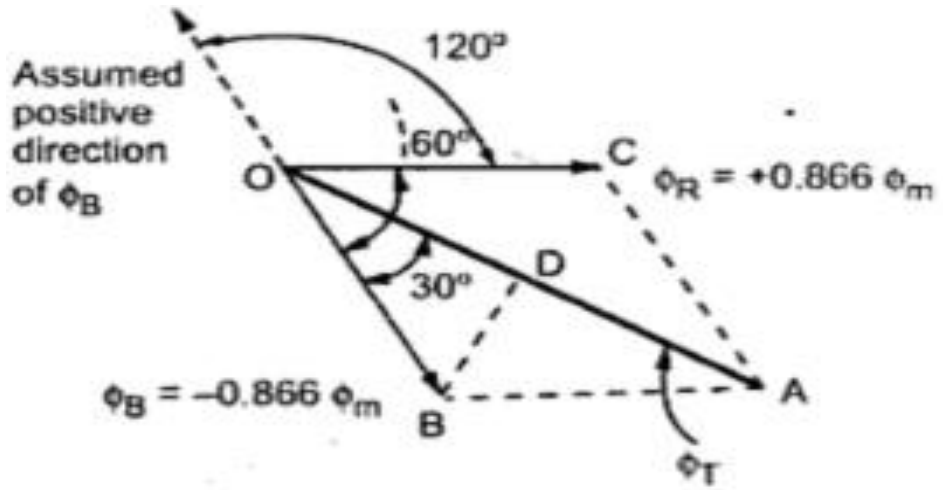
But it can be seen that though its magnitude is 1.5 Φm it has rotated through 60 degrees in space, in clockwise direction, from its previous position.

Case 3: ωt = 120° or θ = 120°

$\Phi_R = \Phi m \sin 120^o = +0.866\ \Phi m$

$\Phi_Y = \Phi m \sin 0^o = 0$

$\Phi_B = \Phi m \sin (-120^o) = -0.866\ \Phi m$



Vector diagram of θ = 120°

After doing the construction same as before i.e. drawing perpendicular from B on Φ_T, it can be provided again that,

$\Phi_T = 1.5\ \Phi m$

But the position of Φ_T is such that it has rotated further through 60° from its previous position, in clockwise direction.

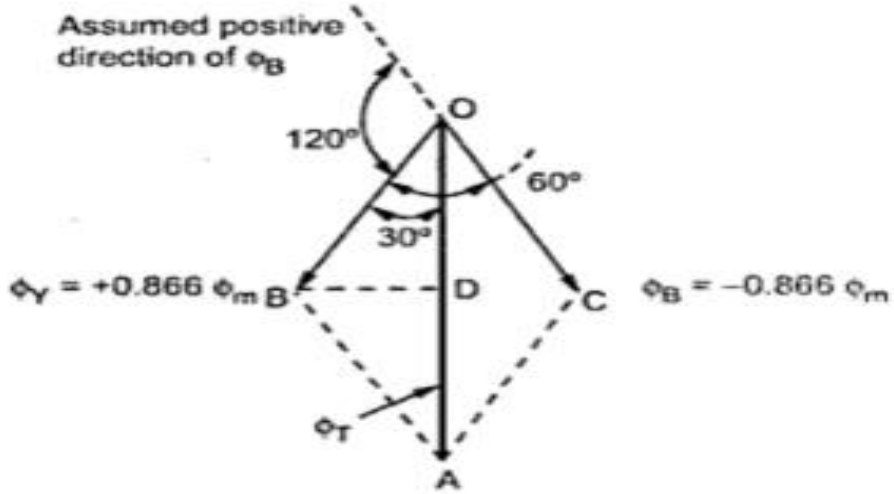
And from its position at θ = 0₀, it has rotated through 120₀ in space, in clockwise direction.

Case 4: ωt = 180° or θ = 180°

$\Phi_R = \Phi m \sin(180^o) = 0$

$\Phi_Y = \Phi m \sin(60^o) = +0.866\ \Phi m$

$\Phi_B = \Phi m \sin(-60^o) = -0.866\ \Phi m$



Vector diagram of θ = 180°

So $\Phi_R = 0$, Φ_Y is positive and Φ_B is negative. Drawing Φ_Y and Φ_B in the appropriate directions, we get the phasor diagram as shown.

From phasor diagram, it can be easily proved that,

$\Phi_T = 1.5\ \Phi m$

Thus the magnitude of Φ_T once again remains same. But it can be seen that it has further rotated through 60_o from its previous position in clockwise direction.

Speed of Rotating Magnetic Field

The speed at which the rotating magnetic field revolves is called the **synchronous speed (N_s)**.

This is the speed which R.M.F rotates in space. In general, for P poles, the rotating field makes one revolution in P/2 cycles of current.

$$\therefore \quad Cycles \quad of \quad Current \quad = \quad \frac{P}{2} \quad \times \quad revolutions \quad of \quad field$$

$$\begin{matrix} or \quad Cycles \quad of \quad current \quad per \quad second \\ revolutions \quad of \quad field \quad per \quad second \end{matrix} \quad = \quad \frac{P}{2} \quad \times \text{ Revolution of field}$$

Since revolutions per second is equal to the revolutions per minute (N s) divided by 60 and the number of cycles per second is the frequency f,

$$f \quad = \quad \frac{P}{2} \times \frac{N_S}{60} \quad = \quad \frac{N_S P}{120}$$

$$\therefore \quad N_S \quad = \quad \frac{120 f}{P}$$

$$N_s = \frac{120 f}{p} = \text{Speed of R.M.F.}$$

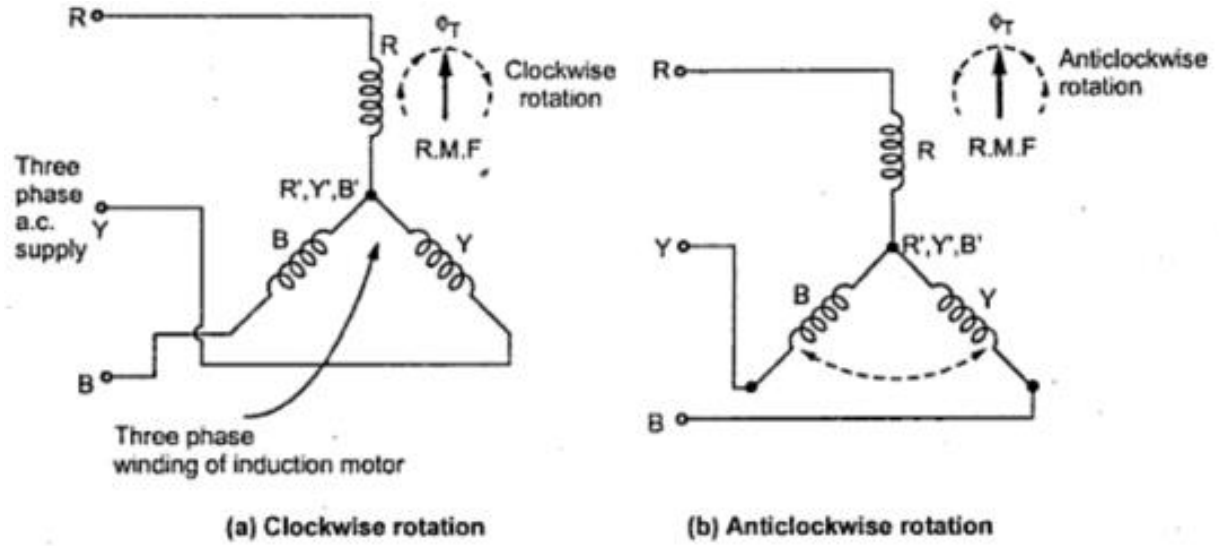
Where f = Supply frequency in Hz

p = Number of poles for which winding is wound

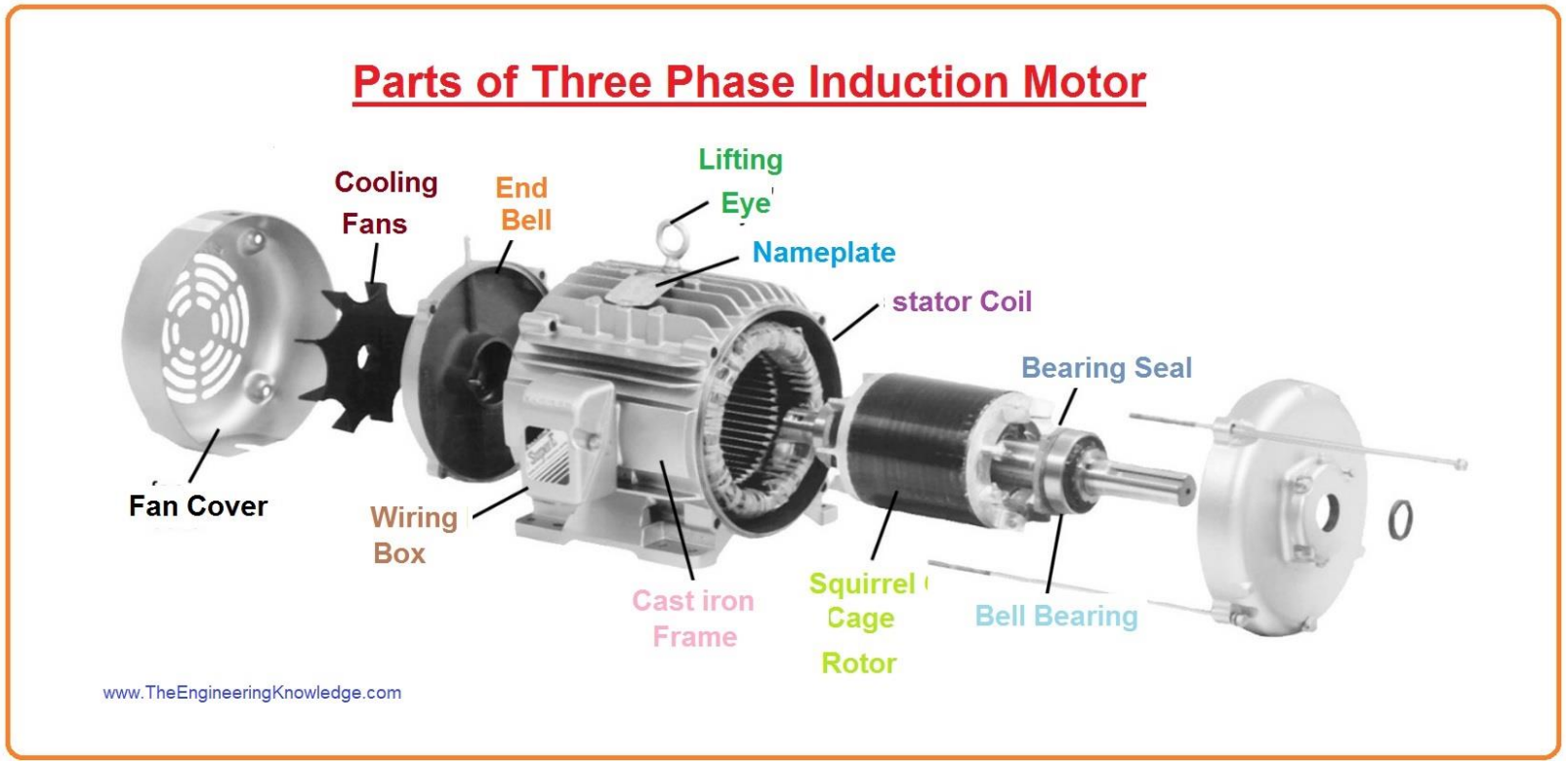
Direction of R.M.F.

- The phase sequence of the three-phase voltage applied to the stator winding in the first figure is R-Y-B.
- If this sequence is changed to R-B-Y, it is observed that the direction of rotation of the field is reversed i.e., the field rotates counterclockwise rather than clockwise.
- However, the number of poles and the speed at which the magnetic field rotates remain unchanged.

Note: Thus by interchanging any two terminals of three phase winding while connecting it to three phase a.c. supply, direction of rotation of R.M.F. gets reversed.



CONSTRUCTION OF THREE PHASE INDUCTION MOTOR

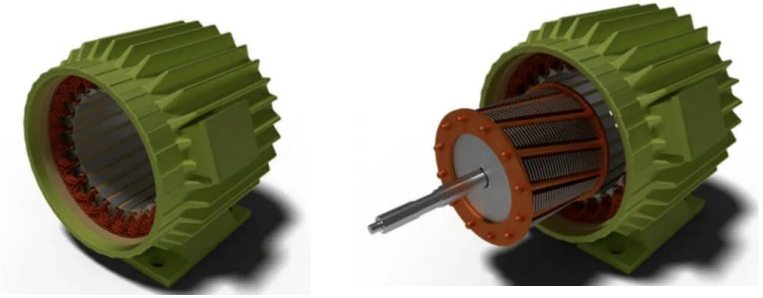


CONSTRUCTION OF THREE PHASE INDUCTION MOTOR

Induction motor has two parts, stator and rotor.

1.The part i.e. three phase windings, which is stationary called stator.

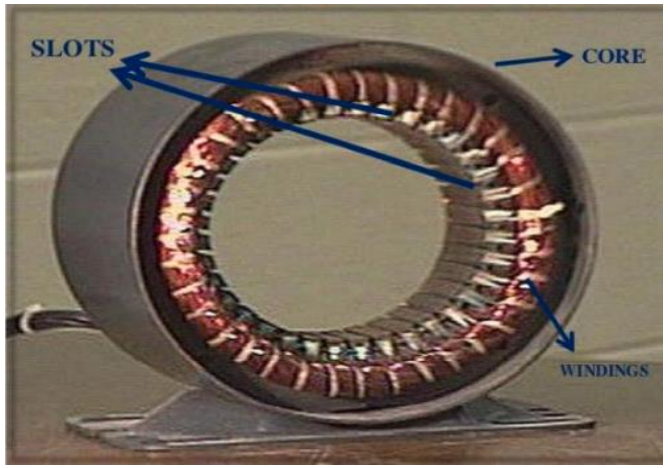
2.The part which rotates and is connected to the mechanical load through shaft called rotor.



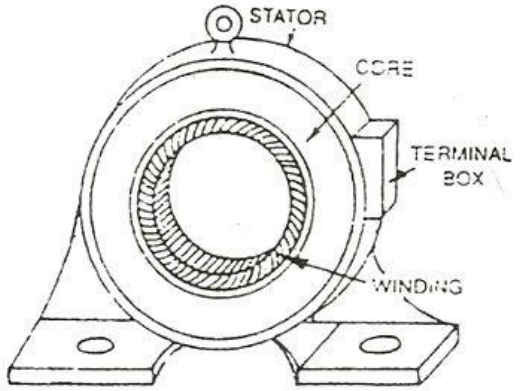
Stator:

The stator is built up of high-grade alloy steel laminations to reduce eddy current losses. It is made up of stampings which are 0.4 to 0.5 mm thick.

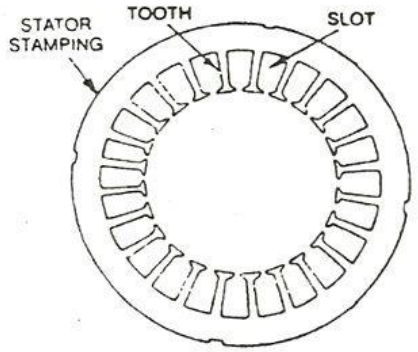
It has three main parts, namely the
1.Outer frame
2..Stator core
3.Stator winding.



Outer frame : It is the outer body of the motor. Its main function is to support the stator core and to protect the inner parts of the machine. For small machines, the outer frame is casted, but for the large machine, it is fabricated.



Stator Core: The stator core is built of high-grade silicon steel stampings. Its main function is to carry the alternating magnetic field which produces hysteresis and eddy current losses. The stampings are fixed to the stator frame. Each stamping is insulated from the other with a thin varnish layer. The thickness of the stamping usually varies from **0.3** to **0.5 mm**.



Stator windings: The stator of the motor is wound for a definite number of poles, depending on the speed of the motor. If the number of poles is greater, the speed of the motor will be less and if the number of poles is less then the speed will be high.

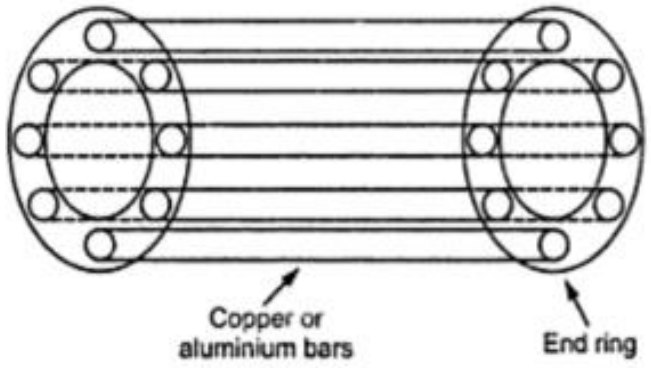
$$N_S \propto \frac{1}{P} \quad \text{or} \quad N_S = \frac{120f}{P}$$

Rotor

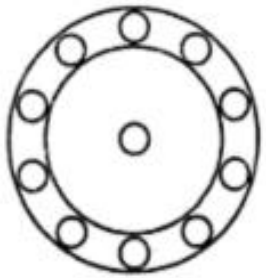
The rotor is placed inside the stator. The rotor core is also laminated in construction and uses cast iron. These are two types.
The laminated cylindrical core is mounted directly on the shaft.

1.Squirrel Cage Rotor

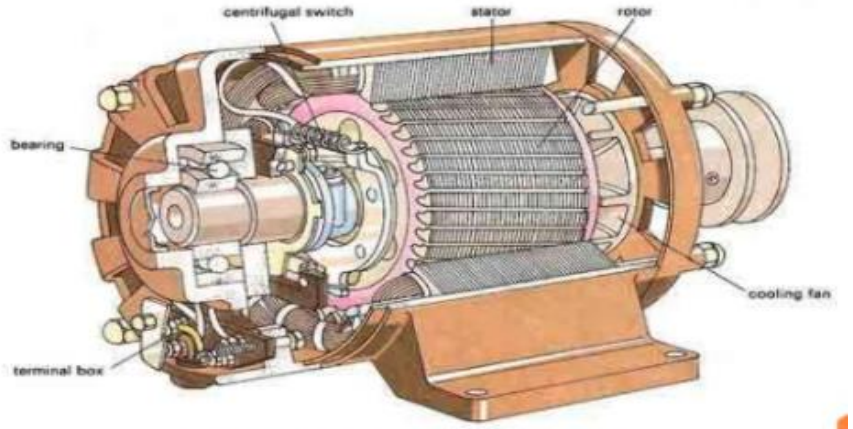
A squirrel cage rotor consists of a laminated cylindrical core. The circular slots at the outer periphery are semi-closed. Each slot contains an uninsulated bar conductor of aluminium or copper. At the end of the rotor the conductors are short-circuited by a heavy ring of copper or aluminium.



(a) Cage type structure of rotor



(b) Symbolic representation



Squirrel Cage Induction Motor

The skewing of the rotor conductors has the following advantages

1. It reduces humming and provides smooth and noise-free operation.
2. It results in a uniform torque curve for different positions of the rotor.
3. The locking tendency of the rotor is reduced. As the teeth of the rotor and the stator attract each other and lock.
4. It increases the rotor resistance due to the increased length of the rotor bar conductors.

Advantages of Squirrel Cage Rotor

The following advantages of the cage rotor are given below:

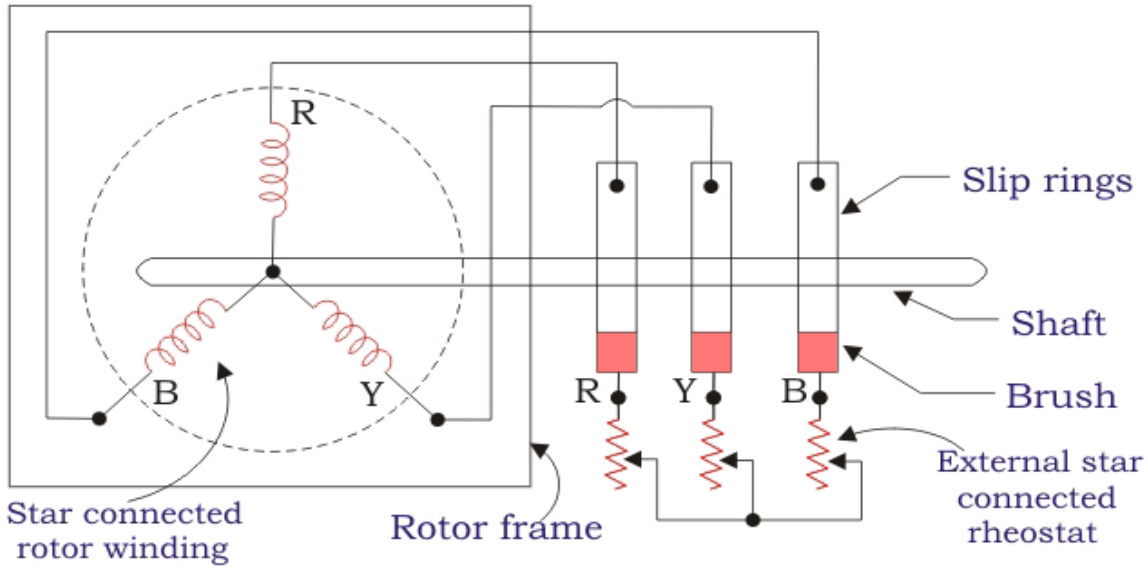
1. The cage rotor is cheaper, and the construction is robust.
2. The absence of the brushes reduces the risk of sparking.
3. Its maintenance is less.
4. The power factor is higher.
5. The efficiency of the cage rotor is higher.

2.Slip ring or Phase Wound Rotor

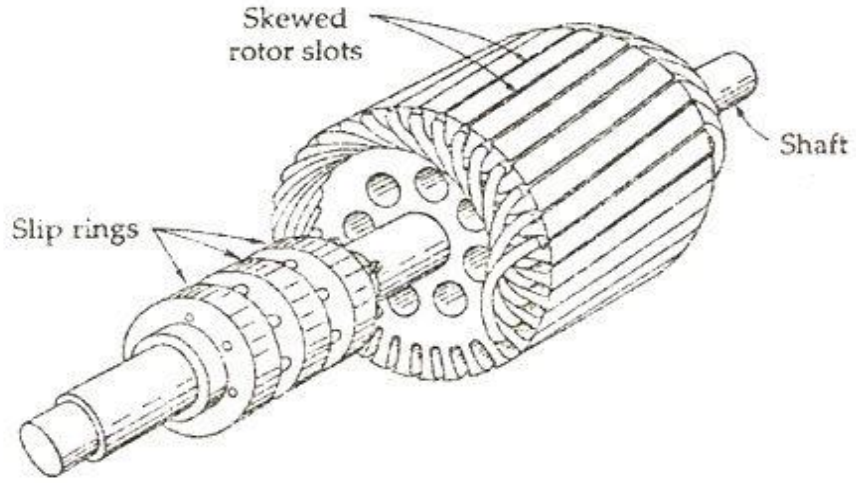
The phase wound rotor is also called a Slip Ring Rotor. It consists of a cylindrical core that is laminated.

The outer periphery of the rotor has a semi-closed slot that carries 3 phase insulated windings. The rotor windings are connected to the star.

The slip rings are mounted on the shaft with brushes resting on them. The brushes are connected to the variable resistor. The function of the slip rings and the brushes is to provide a means of connecting external resistors in the rotor circuit.



Slip Ring Three Phase Induction Motor

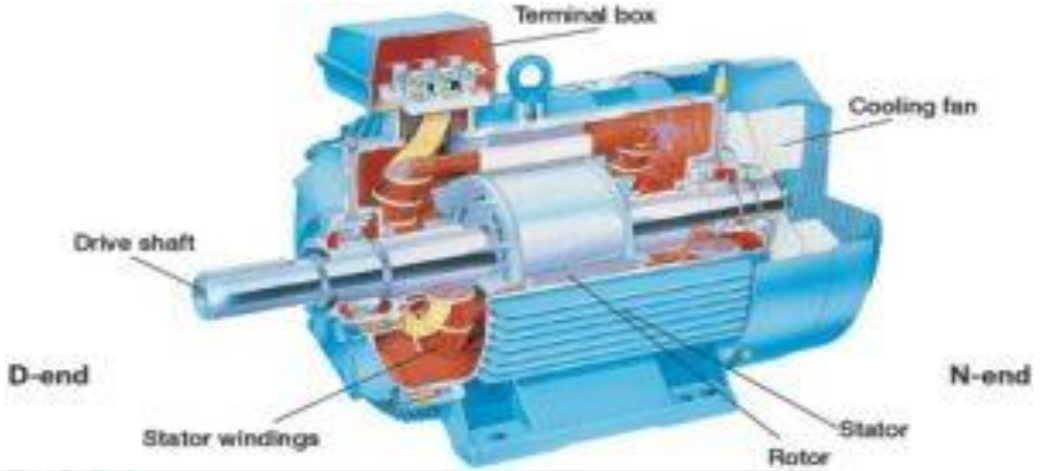


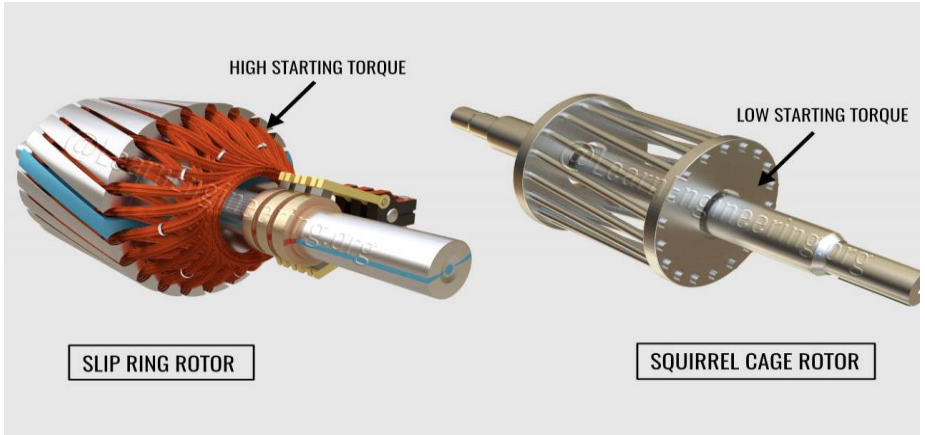
The resistor enables the variation of each rotor phase resistance to serve the following purposes given below:

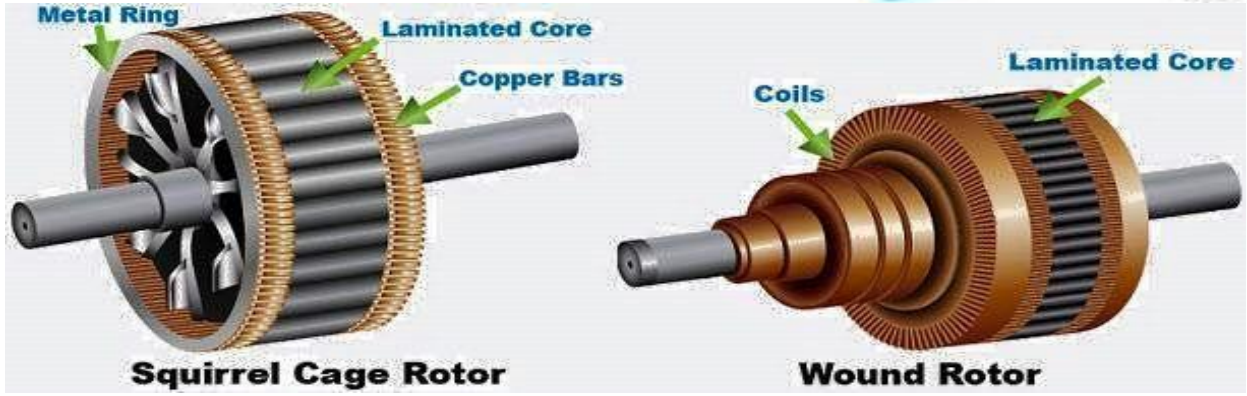
1. It increases the starting torque and decreases the starting current.
2. It is used to control the speed of the motor.
3. In this type also, the rotor is skewed. A mild steel shaft is passed through the center of the rotor and is fixed to it.
4. The purpose of the shaft is to transfer mechanical power.

Advantages of Phase Wound Rotor

1. High starting torque and low starting current.
2. For controlling the speed of the motor, an external resistance can be added in the circuit.







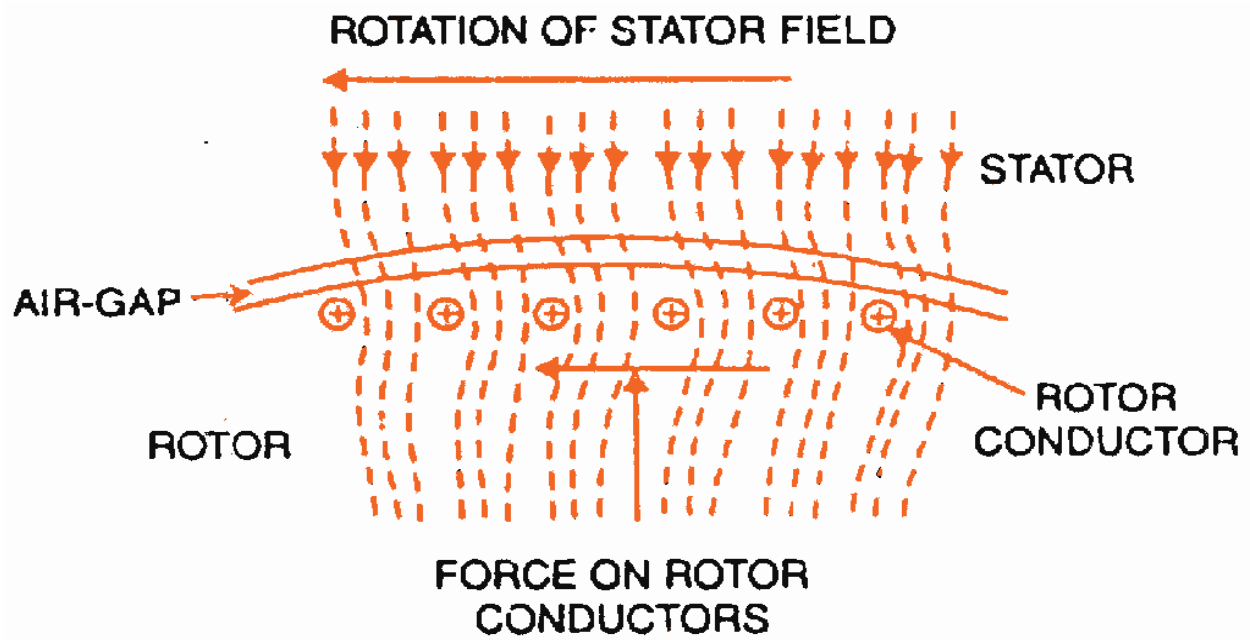
Difference between Slip Ring and Squirrel Cage Induction Motor

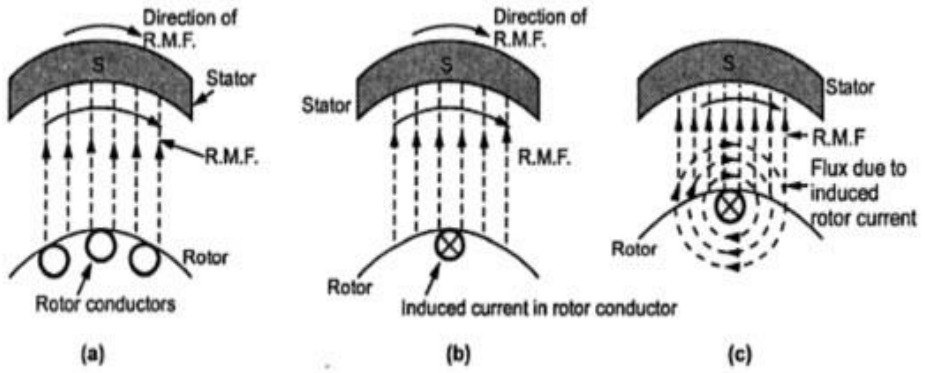
Sr. No.	Wound or slip ring rotor	Squirrel cage rotor
1	Rotor consists of a three phase winding similar to the stator winding.	Rotor consists of bars which are shorted at the ends with the help of end rings.
2	Construction is complicated.	Construction is very simple.
3	Resistance can be added externally.	As permanently shorted, external resistance cannot be added.
4	Slip rings and brushes are present to add external resistance.	Slip rings and brushes are absent.
5	The construction is delicate and due to brushes, frequent maintenance is necessary.	The construction is robust and maintenance free.
6	The rotors are very costly.	Due to simple construction, the rotors are cheap.
7	Only 5% of induction motors in industry use slip ring rotor.	Very common and almost 95% induction motors use this type of rotor.
8	High starting torque can be obtained.	Moderate starting torque which cannot be controlled.
9	Rotor resistance starter can be used.	Rotor resistance starter cannot be used.
10	Rotor must be wound for the same number of poles as that of stator.	The rotor automatically adjusts itself for the same number of poles as that of stator.
11	Speed control by rotor resistance is possible.	Speed control by rotor resistance is not possible.
12	Rotor copper losses are high hence efficiency is less.	Rotor copper losses are less hence have higher efficiency.
13	Used for lifts, hoists, cranes, elevators, compressors etc.	Used for lathes, drilling machines, fans, blowers, water pumps, grinders, printing machines etc.

Working Principle of three-phase induction motor

The motor works on the principle of faraday's law of electromagnetic induction. According to this law, an EMF induced in a conductor due to the rate of change of flux linkage through the conductor.

In squirrel cage induction motor, the rotor bars are short-circuited by end rings and in slip-ring induction motor, the rotor winding is connected with external resistors. And it cut the stator rotating magnetic field





- When three phase stator winding of an induction motor is energized from a 3 phase supply, a rotating magnetic field is set up which rotates round the stator at synchronous speed (Ns = 120 f/P).
- The rotating field passes through the air gap and cuts the rotor conductors, which as yet, are stationary. Due to the relative speed between the rotating flux and the stationary rotor, e.m.f.s are induced in the rotor conductors. Since the rotor circuit is short-circuited, currents start flowing in the rotor conductors.

- The current-carrying rotor conductors are placed in the magnetic field produced by the stator. Consequently, mechanical force acts on the rotor conductors. The sum of the mechanical forces on all the rotor conductors produces a torque which tends to move the rotor in the same direction as the rotating field.

- The fact that rotor is urged to follow the stator field (i.e., rotor moves in the direction of stator field) can be explained by Lenz's law. According to this law, the direction of rotor currents will be such that they tend to oppose the cause producing them.

- The rotor speed (N) of a three phase induction motor is always less than the stator field speed (Ns). This difference in speed depends upon load on the motor. The difference between the synchronous speed Ns of the rotating stator field and the actual rotor speed N is called slip

Formula , $s = \frac{n_s - n_r}{n_s}$

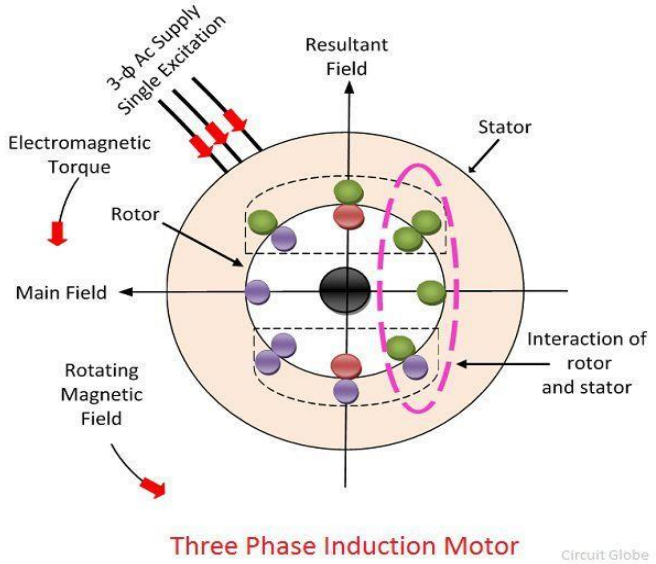
The slip may also be expressed as percent slip as follows :

$\text{Percent slip} = \frac{n_s - n_r}{n_s} \times 100$

- At other speeds, the rotor frequency is proportional to the slip (s): that is,

$f_r = sf$

where f_r - frequency of rotor currents



- When the stator is stationary, Nr=0; so the slip becomes 1 or 100%.
- When Nr is at synchronous speed, the slip becomes zero; so the motor never runs at synchronous speed.
- The slip in the 3 phase induction motor from no load to full load is about 0.1% to 3%; that's why the induction motors are called as constant-speed motors

Alternators/ Synchronous machines

- Electrical machines (generators and motors) operated by alternating current are known as synchronous machines. An alternator is an electromechanical device that converts mechanical energy to electrical energy in the form of alternating current. A.C. generators are also called as alternators.
- A.C. generators are used in to generate electricity in hydroelectric and thermal plants. Alternators are also used in automobiles to generate electricity.
- **The main difference between AC generator and DC generator**
- In DC generator, the armature rotates and the filed system is stationary but in A.C.generator armature is stationary and field system rotates.

And also commutator is present in DC generator , it is not thre in Ac generators.





Construction of Alternator

A.C. generator has mainly two parts

a. Stator

b. Rotor

1. Stator :

- The Stationary part of the alternator is known as stator.
- Armature winding is placed on the stator.The stator consists of a cast iron frame, which suppliels the armature core having slots on its inner periphery for housing the armature conductors.
- The stator core is built up of sheet steel laminations to reduce the eddy current losses
- Ventilating ducts are provided in the frame for cooling the machine.
- The slots are open, semi closed or closed type.



2.Rotor:

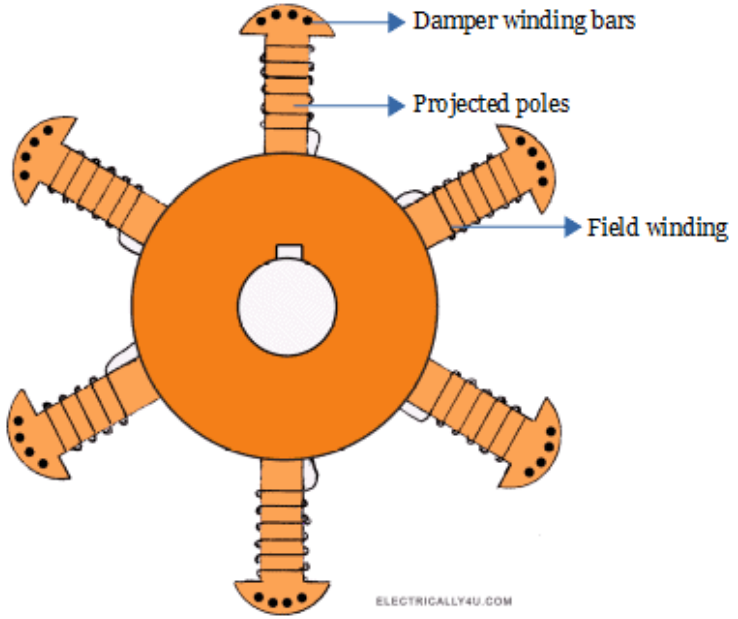
- It is a rotating part of a machine. Rotor carries a field winding which is supplied with direct current through two sliprings by DC source.
- This DC sources is known as exciter is generally a small dc shunt or compound generator mounted on the shaft of the alternator itself.
- Rotors are two types

(a) Salient pole type(or Projecting pole type)
(b) Non-salient pole type(or smooth cylindrical type)



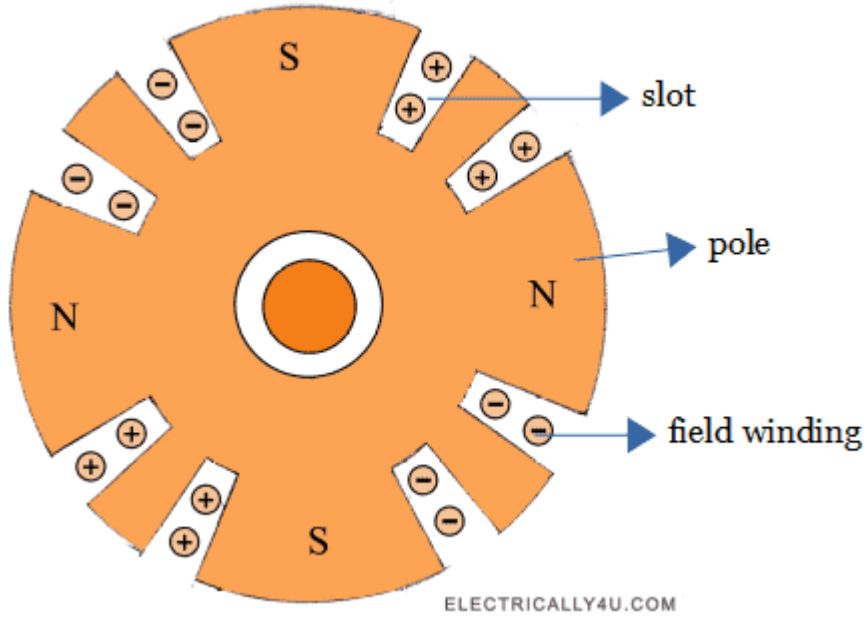
(a) Salient pole type(or Projecting pole type) :

- This type of rotor is used for low to medium speed/rpm alternators, were more number of poles are required may be 20 or 30 poles. The poles are made of thick laminated steel sections pivoted together.
- Such alternator can be identified by large diameter and short axial length.
- The pole is made of steel or cast iron and the pole winding is excited by a D.C. generator driven by the shaft of alternator.



(b) Non-salient pole type(or smooth cylindrical type)

- In this rotors are used for high speed alternators for turbo alternators.
- It has small diameter and very long axial length.
- The rotor is made of steel cylinder with number of slots cut on the periphery of the cylinder.
- The rotors are generally designed for 2(3000rpm) or 4 poles(1500 rmp)
- These alternators usually not provided with damper winding since the solid field poles present will themselves act as efficient dampers.





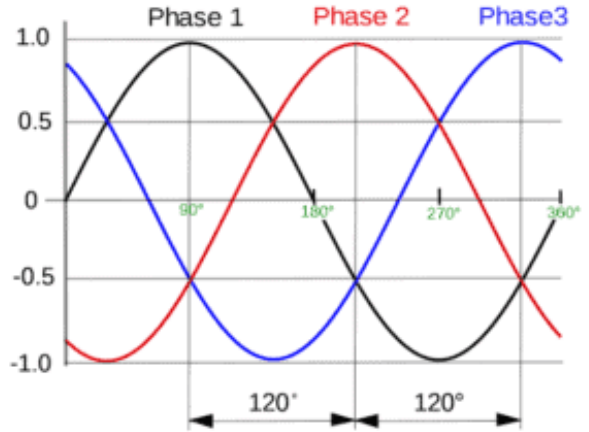


Sr. No.	Salient Pole Type	Smooth Cylindrical Type
1.	Poles are projecting out from the surface.	Unslotted portion of the cylinder acts as poles hence poles are non-projecting.
2.	Air gap is non-uniform.	Air gap is uniform due to smooth cylindrical periphery.
3.	Diameter is high and axial length is small.	Small diameter and large axial length is the feature.
4.	Mechanically weak.	Mechanically robust.
5.	Preferred for low speed alternators.	Preferred for high speed alternators i.e. for turboalternators.
6.	Prime mover used are water turbines, I.C. engines.	Prime movers used are steam turbines, electric motors.
7.	For same size, the rating is smaller than cylindrical type.	For same size, rating is higher than salient pole type.
8.	Separate damper winding is provided.	Separate damper winding is not necessary.

Working of Alternator

The mechanical energy to the alternator is given through the prime mover. Thus the rotor and hence the field magnets are rotated in the direction of prime mover.

The *field poles are magnetized by applying a voltage* to the field coils through slip rings. The field windings are connected such that, alternate North and South poles are produced.



As the rotor rotates, the *North and South poles of the magnetic flux is cut by the stationary armature*, which induces an emf in the armature windings.
As the magnetic poles are alternately North and South Poles, the emf thus generated will be alternating in nature.

The frequency of the induced em depends on the number of N and S poles moving past the armature conductor in one second. The direction of induced emf can be found by Fleming's right hand rule and frequency in given by

$$f = \frac{PN}{120}$$

Difference between a Generator and an Alternator

Sl. No.	Differentiating Property	Alternator	Generator
1	Definition	An alternator is a device that converts mechanical energy into AC electrical energy.	A generator is a mechanical device which converts mechanical energy to either AC or DC electrical energy.
2	Output Current	An alternator always induces an alternating current.	A generator can generate either alternating or direct current.
3	Energy Efficiency	Alternators are very efficient.	Generators are considered less efficient.
4	Output	Alternators have a higher output than generators.	Generators have a lower output when compared with an alternator.
5	Energy Conservation	Alternators use only the required amount of energy and thus, it conserves more energy.	Generators use all the energy that is produced and so, they conserve less energy.

6	Polarization After Installation	Polarization is not required in case of alternators.	Generators need to be polarized after installation.
7	Magnetic Field	The magnetic field is rotating inside the stator of an alternator.	In case of a generator, the magnetic field is stationary or fixed where the armature winding spins.
8	Armature Movement	The armature of an alternator is stationary.	The armature of a generator is rotating.
9	Input Supply	The alternator takes input supply from the stator.	The generator takes input supply from the rotor.
10	RPM (Rotation Per Minute) Range	Alternators have a wide range of RPM.	Generators have a low range of RPM.
11	Voltage Generation	Alternators produce voltage only when needed.	Generators produce voltage at all times.
12	Size	Alternators are generally smaller in size.	Generators are larger and require more space to fit in.
13	Brush Efficiency	The brushes of alternators last longer.	The brushes of generators last lesser when compared to an alternator.
14	Charging of a Dead Battery	Alternators will never charge a dead battery	A generator can be used for charging a dead battery.
15	Uses	Alternators are mainly used in the automobile industry as a charging system for the battery.	Generators are widely used to produce large-scale electricity.

E.M.F. equation of Alternator

Let Φ = Flux per pole, in Wb
P = Number of poles
N = Synchronous speed in r.p.m
f = Frequency of induced emf in Hz
Z = Total number of conductors
Zph = Conductors per phase connected in series
Zph = Z/3 as number of phases = 3

$$K_f = \frac{\text{R.M.S.}}{\text{Average}} = 1.11$$

$\therefore$ R.M.S. value of E_{ph} = K_f × Average value

$$\therefore \quad E_{ph} = 1.11 \times 4\, f\, \phi\, T_{ph}$$

$$\boxed{E_{ph} = 4.44\, f\, \phi\, T_{ph} \text{ volts}}$$

Applications,Advantages and Disadvantages

Applications

- Automobiles
- Electrical power generator plants
- Marine applications
- Diesel electrical multiple units
- Radiofrequency transmission

Advantages

- Cheap
- Low weight
- Low maintenance
- Construction is simple
- Robust
- More compact

Disadvantages

- Alternators need transformers
- Alternators will overheat if the current is high

Summary of the electrical machines….

- Generation of rotating magnetic fields, Construction and working of a three-phase induction motor.
- Significance of torque-slip characteristic.
- Loss components and efficiency, starting and speed control of induction motor.
- Single-phase induction motor. Construction, working
- Torque-speed characteristic and speed control of separately excited dc motor.
- Construction and working of synchronous generators.

Thank you